Signals and Systems

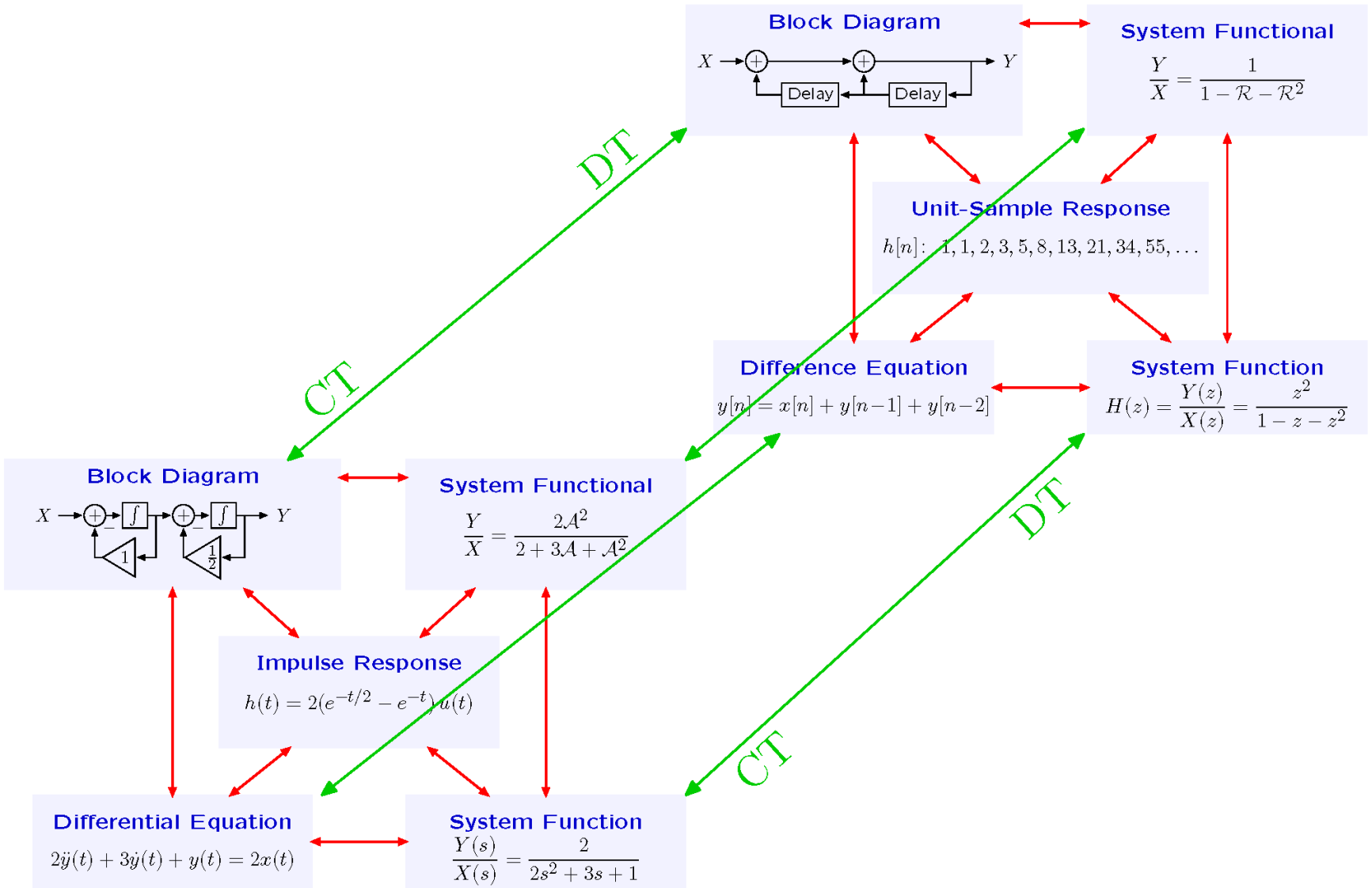
Lecture 10: Continuous-time Fourier Series

Instructor: Prof. Ting Zhang
Zhejiang University

04/15/2025
Adapted from the materials provided on
the MIT OpenCourseWare

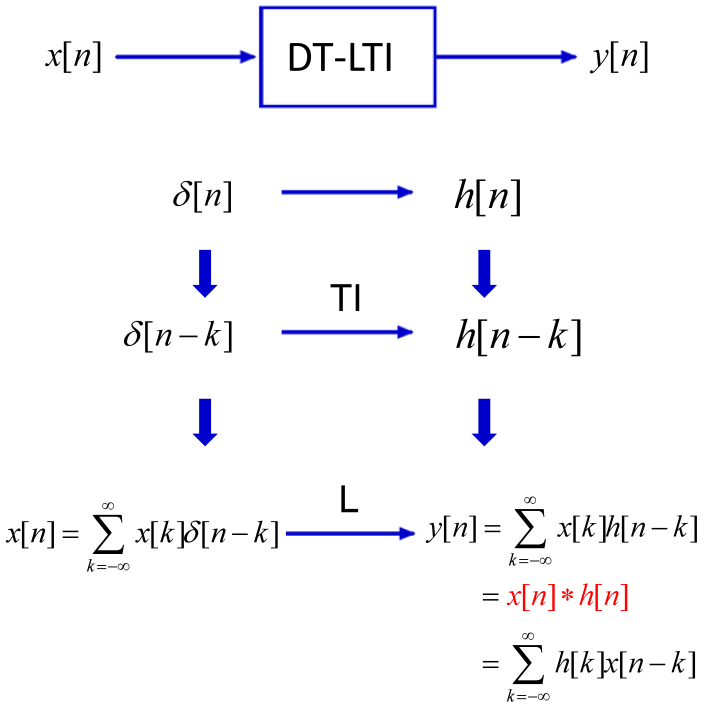
Review: Concept Maps

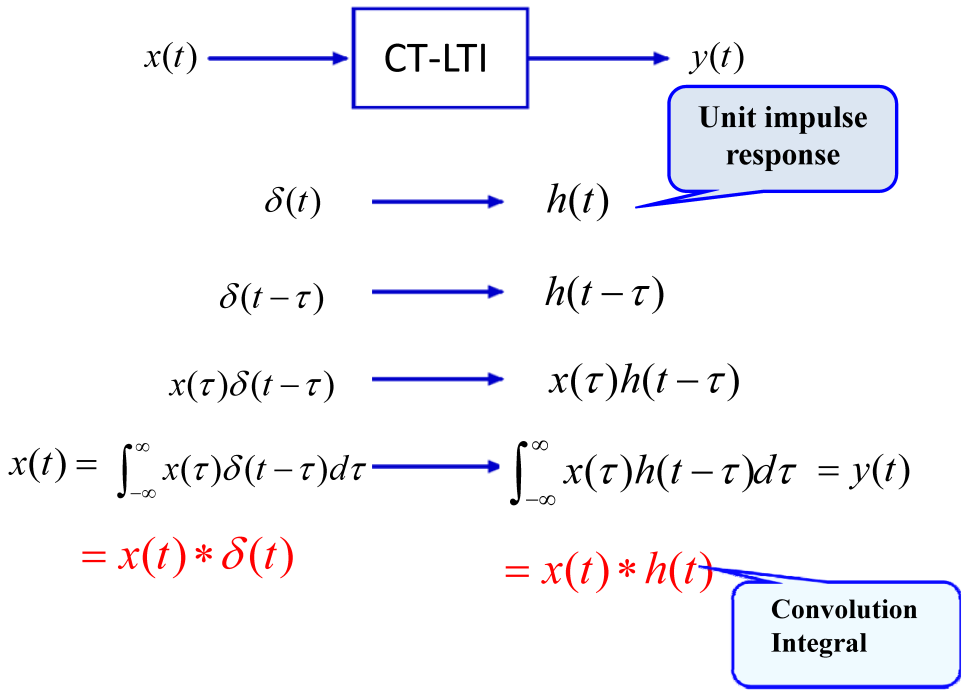
Relations between CT and DT representations.



Review: Building-block Signals

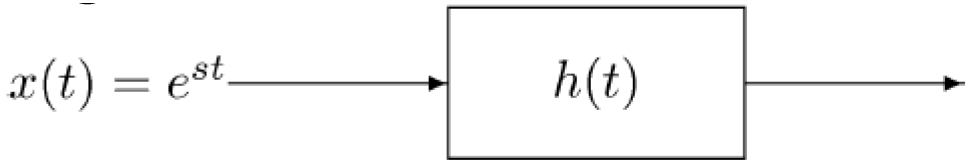
- Unit impulse function



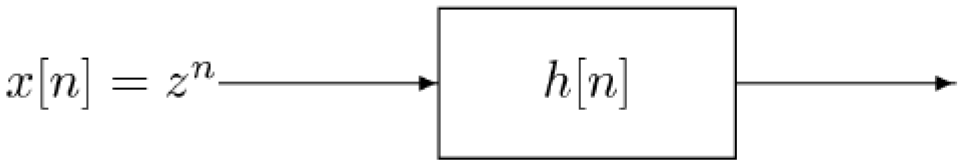


Review: Building-block Signals

- Complex exponential function



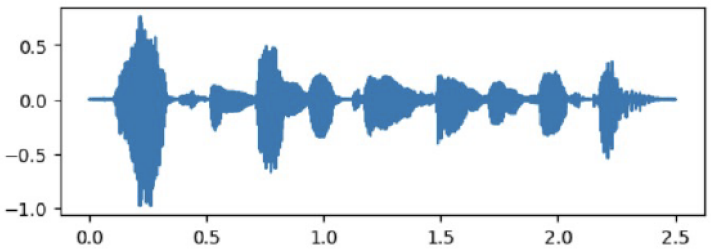
$$
\begin{aligned}
y(t) &= \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau \\
&= \left[\int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau \right] e^{st} \\
&= \underbrace{H(s)}_{\text{eigenvalue}} \underbrace{e^{st}}_{\text{eigenfunction}}
\end{aligned}
$$

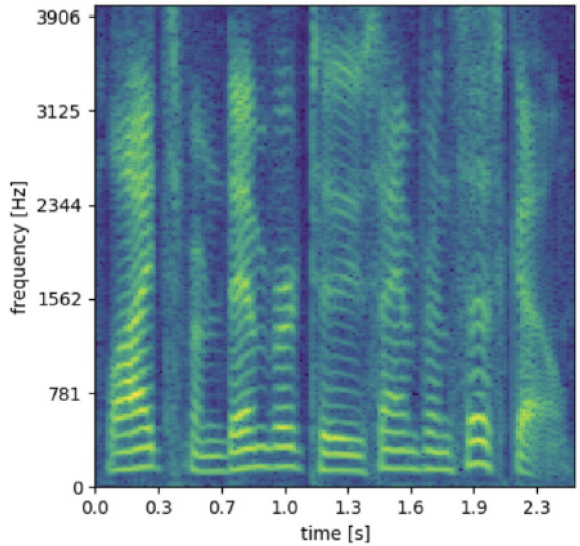


$$
\begin{aligned}
y[n] &= \sum_{m=-\infty}^{\infty} h[m] z^{n-m} \\
&= \left[\sum_{m=-\infty}^{\infty} h[m] z^{-m} \right] z^n \\
&= \underbrace{H(z)}_{\text{eigenvalue}} \underbrace{z^n}_{\text{eigenfunction}}
\end{aligned}
$$

Fourier Representations

- Multiple different ways of looking at a signal.
  - E.g. Two representations of a speech signal:





- We will primarily focus on a family of transforms referred to as Fourier transform;
- End goal: Fourier representation of arbitrary signals;
- Simplest case: representing a periodic signal as sums of sinusoids (Harmonics);

Harmonics

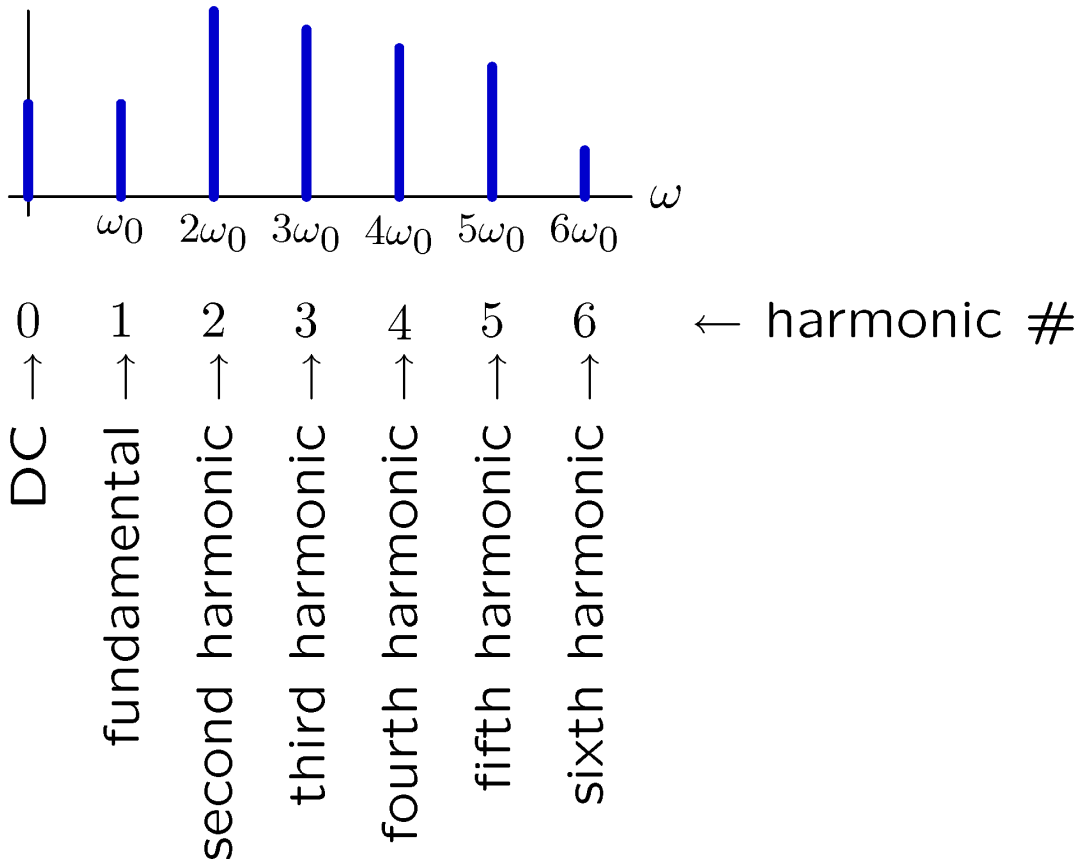
A harmonically related set of complex exponentials is a set of periodic exponentials with fundamental frequencies that are all multiples of a single positive frequency ω_0:

$$\phi_k(t) = e^{jk\omega_0 t}, \quad k = 0, \pm 1, \pm 2, ...$$

- $k = 0$, DC component;
- $k = \pm 1$, the first harmonic components;
- $k = \pm 2$, the second harmonic components;
- ...

Harmonics

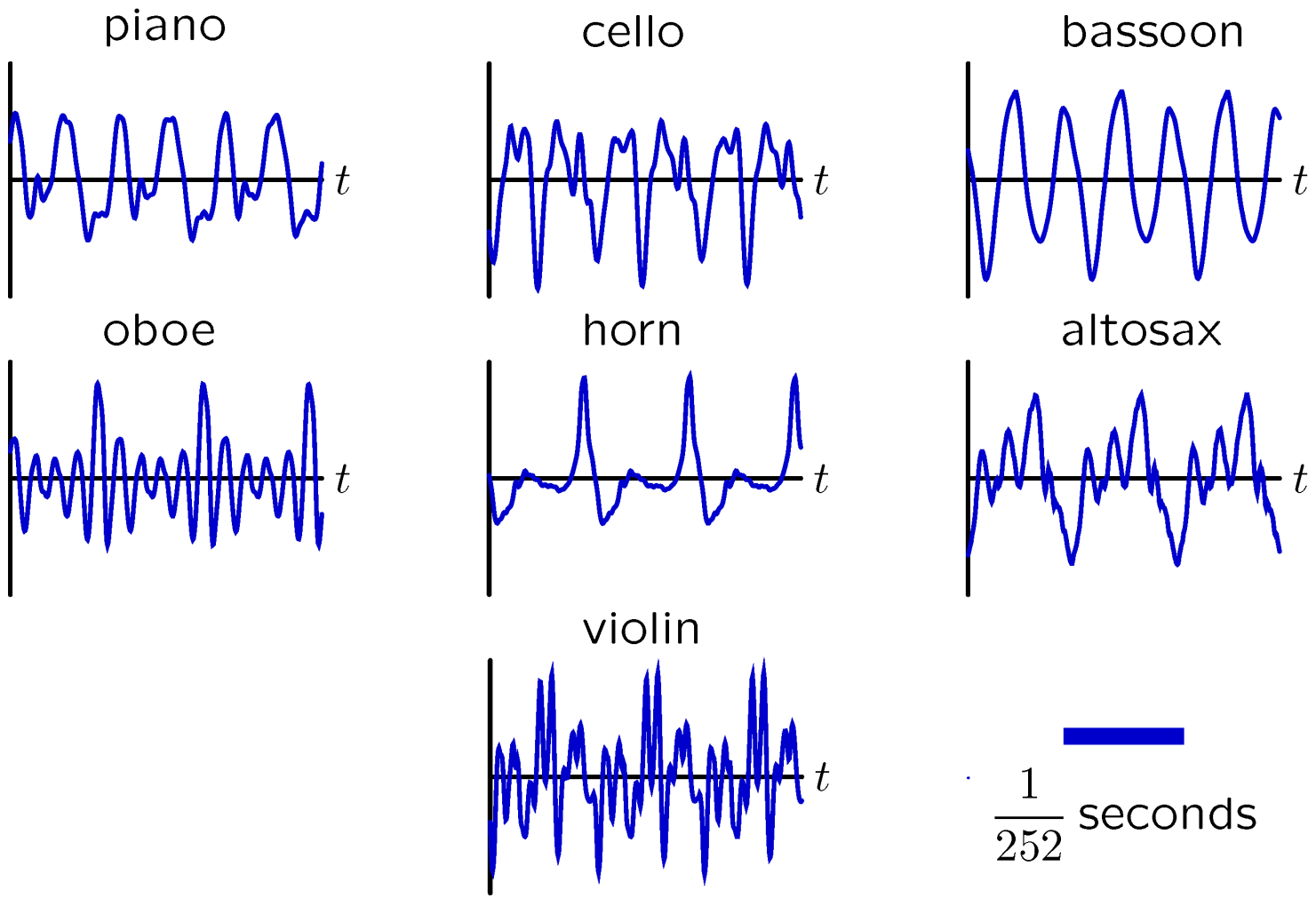
Representing signals by their harmonic components.



Musical Instruments

Harmonic content is natural way to describe some kinds of signals.

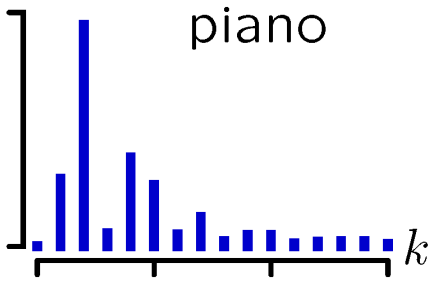
Ex: musical instruments (http://theremin.music.uiowa.edu/MIS)

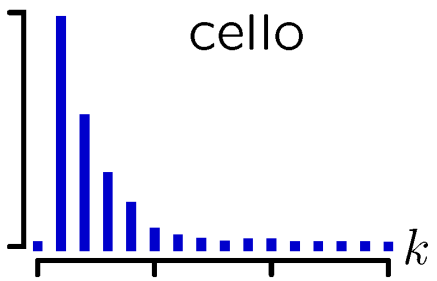


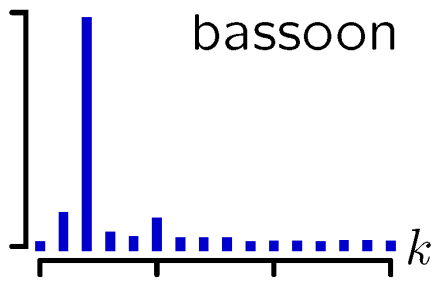
Musical Instruments

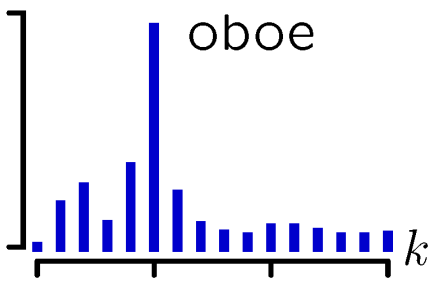
Harmonic content is natural way to describe some kinds of signals.

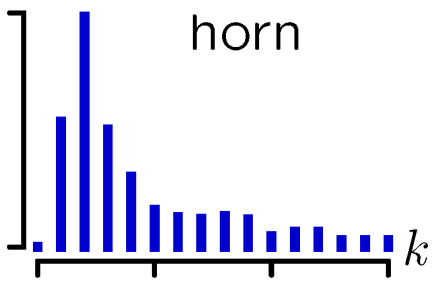
Ex: musical instruments

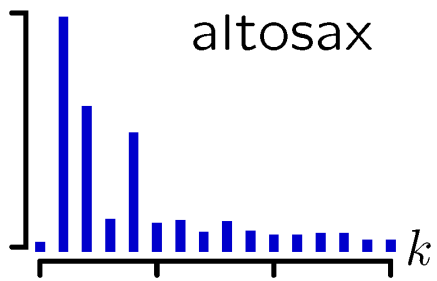


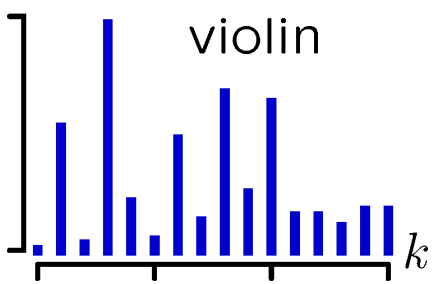






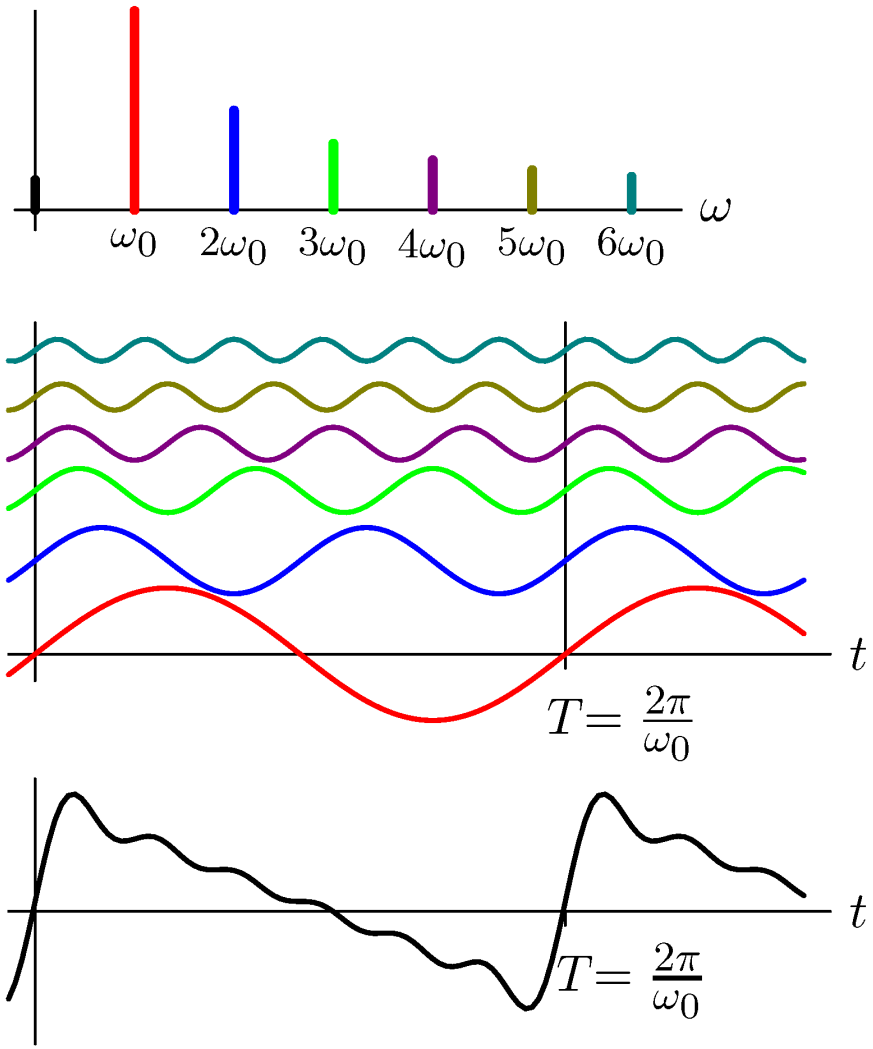






Harmonic Representations

What signals can be represented by sums of harmonic components?



Only periodic signals: all harmonics of ω_0 are periodic in $T = 2\pi/\omega_0$.

Underlying properties of harmonic components

Underlying properties.

1. Multiplying two harmonics produces a new harmonic with the same fundamental frequency:

$$e^{jk\omega_0 t} \times e^{jl\omega_0 t} = e^{j(k+l)\omega_0 t}.$$

2. The integral of a harmonic over any time interval with length equal to a period T is zero unless the harmonic is at DC:

$$\int_{t_0}^{t_0+T} e^{jk\omega_0 t} dt \equiv \int_T e^{jk\omega_0 t} dt = \begin{cases} 0, & k \neq 0 \\ T, & k = 0 \end{cases}$$
$$= T\delta[k]$$

How to find the coefficients

Assume that $x(t)$ is periodic in T and is composed of a weighted sum of harmonics of $\omega_0 = 2\pi/T$.

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\omega_0 kt}$$

Then

$$\begin{aligned}\int_T x(t)e^{-jl\omega_0 t}dt &= \int_T \sum_{k=-\infty}^{\infty} a_k e^{j\omega_0 kt} e^{-j\omega_0 lt}dt \\ &= \sum_{k=-\infty}^{\infty} a_k \int_T e^{j\omega_0(k-l)t}dt \\ &= \sum_{k=-\infty}^{\infty} a_k T\delta[k-l] = Ta_l\end{aligned}$$

Therefore

$$a_k = \frac{1}{T}\int_T x(t)e^{-j\omega_0 kt}dt \qquad = \frac{1}{T}\int_T x(t)e^{-j\frac{2\pi}{T}kt}dt$$

Fourier Series

Determining harmonic components of a periodic signal.

$$a_k = \frac{1}{T}\int_T x(t)e^{-j\frac{2\pi}{T}kt}dt \qquad \text{(“analysis” equation)}$$

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt} \qquad \text{(“synthesis” equation)}$$

Check Yourself

Determine the Fourier series representation for

$$x(t) = \cos 4\pi t + 2\sin 8\pi t$$

Check Yourself

Determine the Fourier series representation for

$$x(t) = \cos 4\pi t + 2\sin 8\pi t$$

Euler's relation (memorize!)

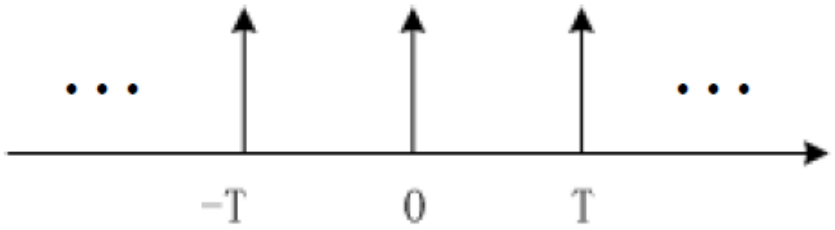
$$= \frac{1}{2}\left[e^{j4\pi t} + e^{-j4\pi t}\right] + \frac{2}{2j}\left[e^{j8\pi t} - e^{-j8\pi t}\right]$$

$$\omega_0 = 4\pi \qquad T = \frac{2\pi}{\omega_0} = \frac{2\pi}{4\pi} = \frac{1}{2}$$

$$\begin{aligned}
a_0 &= 0 \\
a_1 &= \frac{1}{2} \\
a_{-1} &= \frac{1}{2} \\
a_2 &= \frac{1}{j} \\
a_{-2} &= -\frac{1}{j} \\
a_3 &= 0 \\
a_{-3} &= 0
\end{aligned}$$

Check Yourself

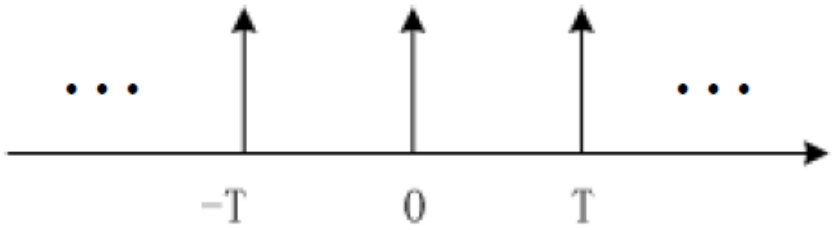
Determine the Fourier series representation for



$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

Check Yourself

Determine the Fourier series representation for



$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} x(t)e^{-jk\omega_0 t}dt = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t)e^{-jk\omega_0 t}dt$$

$$= \frac{1}{T} \quad \text{for } \textit{all } k \text{ !}$$

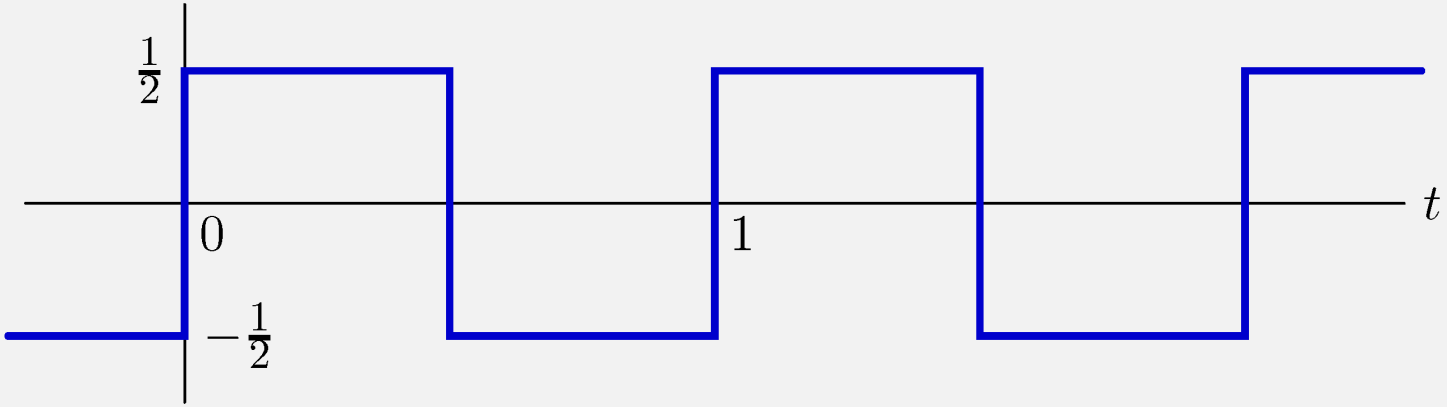
$$\Downarrow$$

$$x(t) = \frac{1}{T}\sum_{k=-\infty}^{\infty} e^{jk\omega_0 t}$$

— All components have:
(1) the same amplitude,
&
(2) the same phase.

Check Yourself

Let a_k represent the Fourier series coefficients of the following square wave.

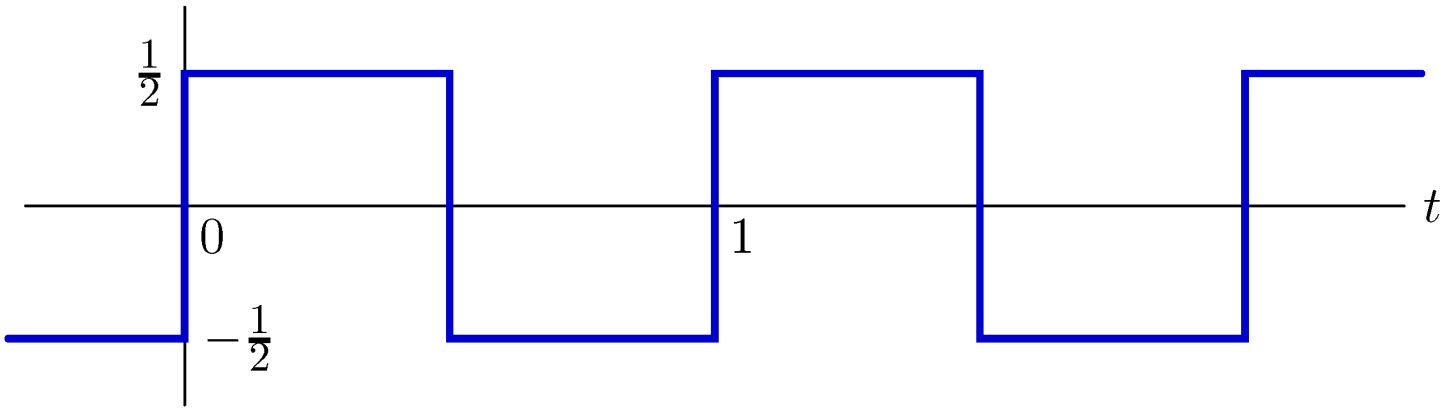


How many of the following statements are true?

1. $a_k = 0$ if k is even
2. a_k is real-valued
3. $|a_k|$ decreases with k^2
4. there are an infinite number of non-zero a_k
5. all of the above

Check Yourself

Let a_k represent the Fourier series coefficients of the following square wave.



$$a_k = \int_T x(t) e^{-j\frac{2\pi}{T}kt} dt = -\frac{1}{2}\int_{-\frac{1}{2}}^{0} e^{-j2\pi kt} dt + \frac{1}{2}\int_{0}^{\frac{1}{2}} e^{-j2\pi kt} dt$$

$$= \frac{1}{j4\pi k}\left(2 - e^{j\pi k} - e^{-j\pi k}\right)$$

$$= \begin{cases} \frac{1}{j\pi k} \; ; & \text{if } k \text{ is odd} \\ 0 \; ; & \text{otherwise} \end{cases}$$

Check Yourself

Let a_k represent the Fourier series coefficients of the following square wave.

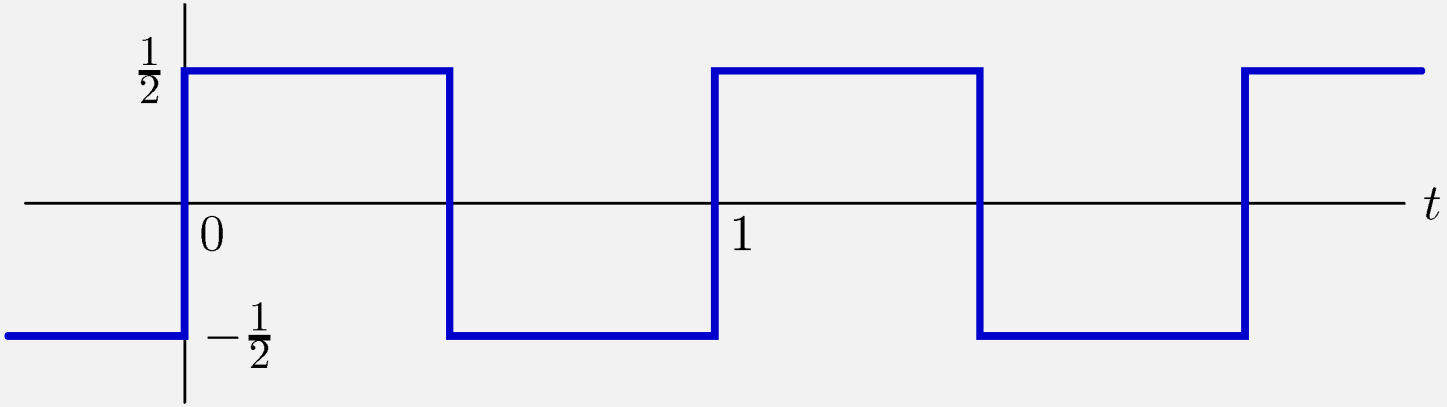
$$a_k = \begin{cases} \frac{1}{j\pi k} \; ; & \text{if } k \text{ is odd} \\ 0 \; ; & \text{otherwise} \end{cases}$$

How many of the following statements are true?

1. $a_k = 0$ if k is even ✓
2. a_k is real-valued ✗
3. $|a_k|$ decreases with k^2 ✗
4. there are an infinite number of non-zero a_k ✓
5. all of the above ✗

Check Yourself

Let a_k represent the Fourier series coefficients of the following square wave.

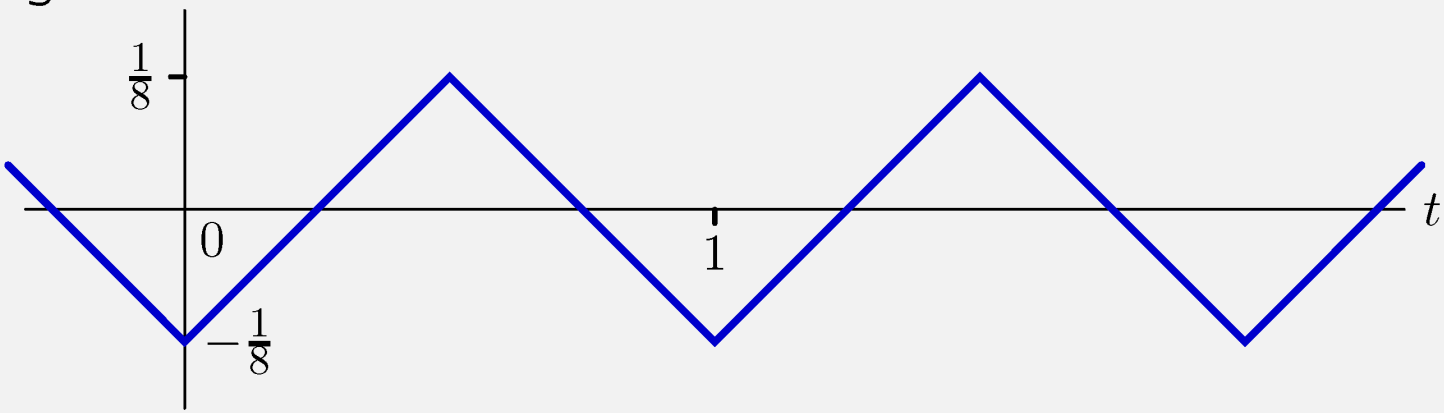


How many of the following statements are true? 2

1. $a_k = 0$ if k is even ✓
2. a_k is real-valued ✗
3. $|a_k|$ decreases with k^2 ✗
4. there are an infinite number of non-zero a_k ✓
5. all of the above ✗

Check Yourself

Let b_k represent the Fourier series coefficients of the following triangle wave.



How many of the following statements are true?

1. $b_k = 0$ if k is even
2. b_k is real-valued
3. $|b_k|$ decreases with k^2
4. there are an infinite number of non-zero b_k
5. all of the above

Fourier Series Properties

If a signal is differentiated in time, its Fourier coefficients are multiplied by $j\frac{2\pi}{T}k$.

Proof: Let

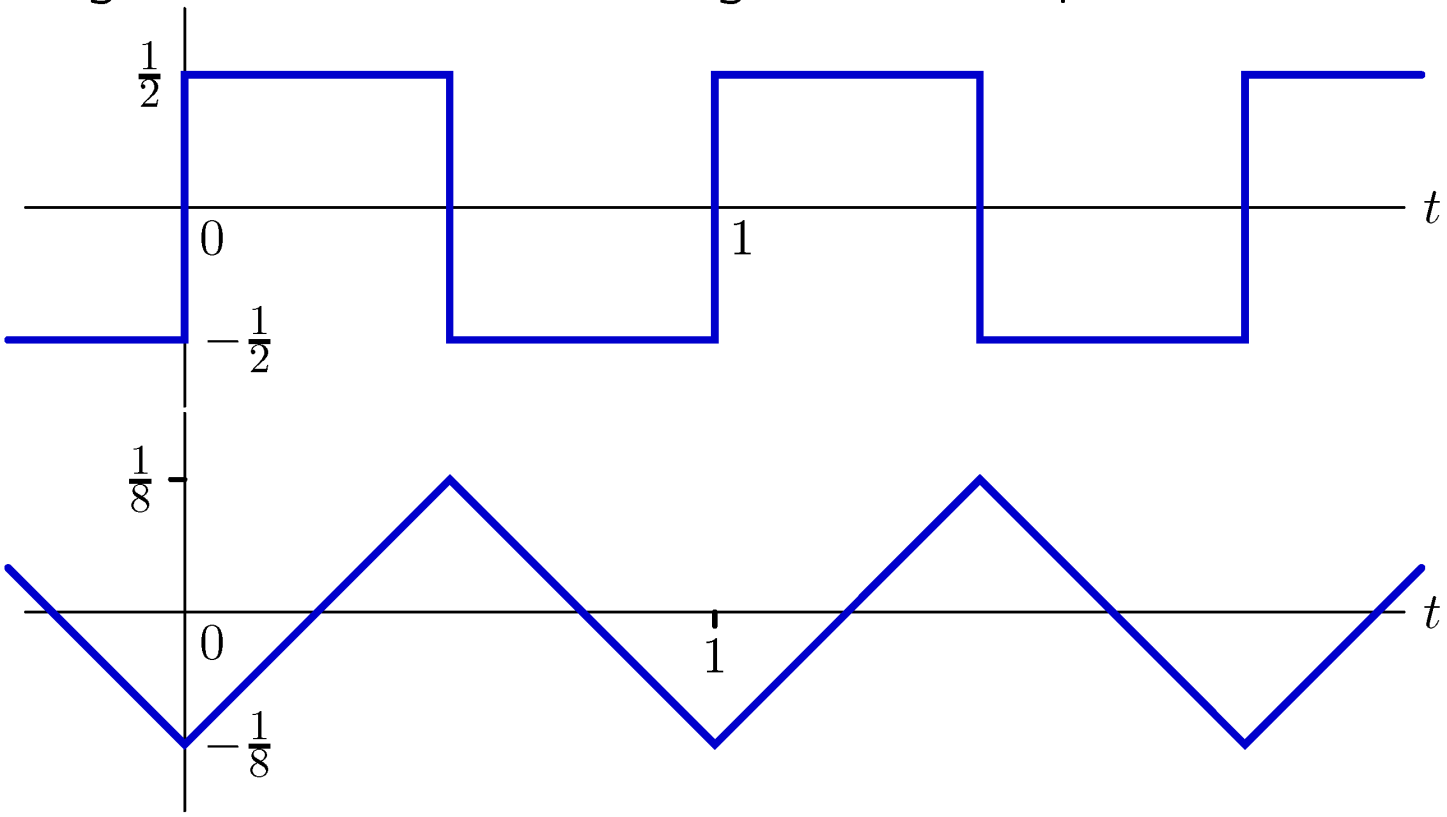
$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt}$$

then

$$\dot{x}(t) = \dot{x}(t+T) = \sum_{k=-\infty}^{\infty} \left(j\frac{2\pi}{T} k\, a_k \right) e^{j\frac{2\pi}{T}kt}$$

Check Yourself

The triangle waveform is the integral of the square wave.



Therefore the Fourier coefficients of the triangle waveform are $\frac{1}{j2\pi k}$ times those of the square wave.

$$b_k = \frac{1}{jk\pi} \times \frac{1}{j2\pi k} = \frac{-1}{2k^2\pi^2} \;;\; k \text{ odd}$$

Check Yourself

Let b_k represent the Fourier series coefficients of the following triangle wave.

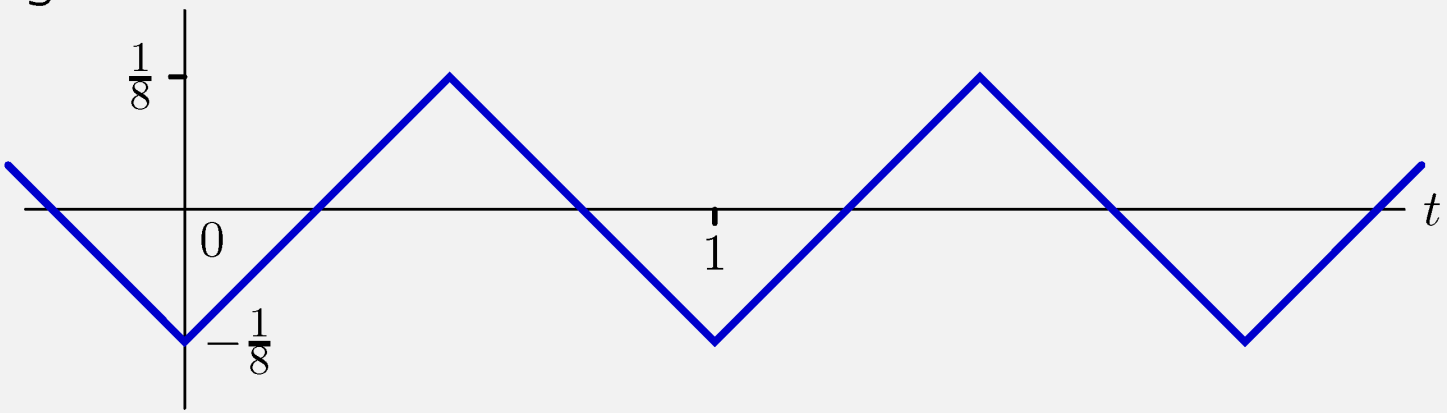
$$b_k = \frac{-1}{2k^2\pi^2} \; ; \; k \text{ odd}$$

How many of the following statements are true?

1. $b_k = 0$ if k is even ✓
2. b_k is real-valued ✓
3. $|b_k|$ decreases with k^2 ✓
4. there are an infinite number of non-zero b_k ✓
5. all of the above ✓

Check Yourself

Let b_k represent the Fourier series coefficients of the following triangle wave.



How many of the following statements are true? 5

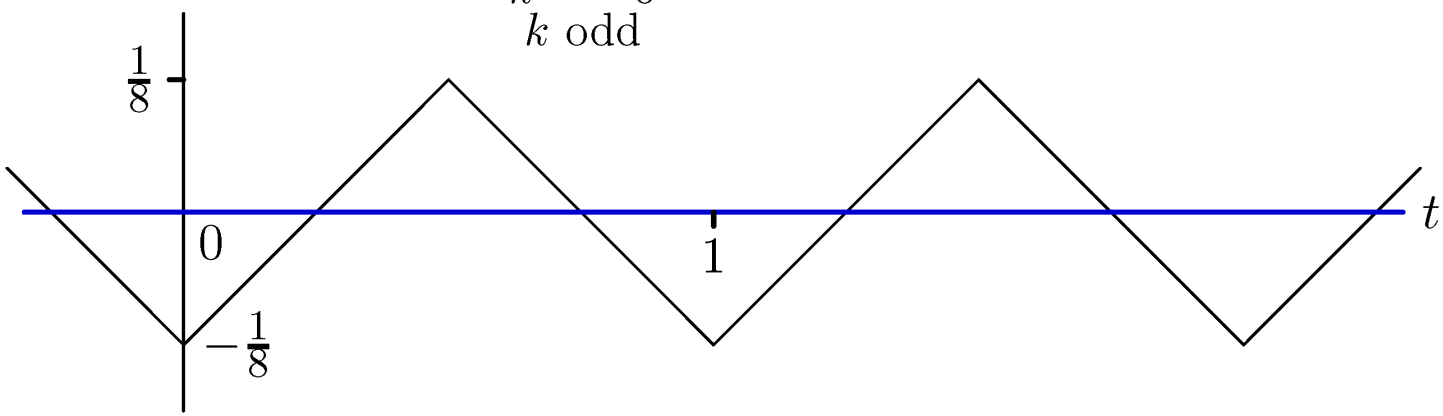
1. $b_k = 0$ if k is even √
2. b_k is real-valued √
3. $|b_k|$ decreases with k^2 √
4. there are an infinite number of non-zero b_k √
5. all of the above √

Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

Example: triangle waveform

$$\sum_{\substack{k=-0 \\ k \text{ odd}}}^{0} \frac{-1}{2k^2\pi^2} e^{j2\pi kt}$$

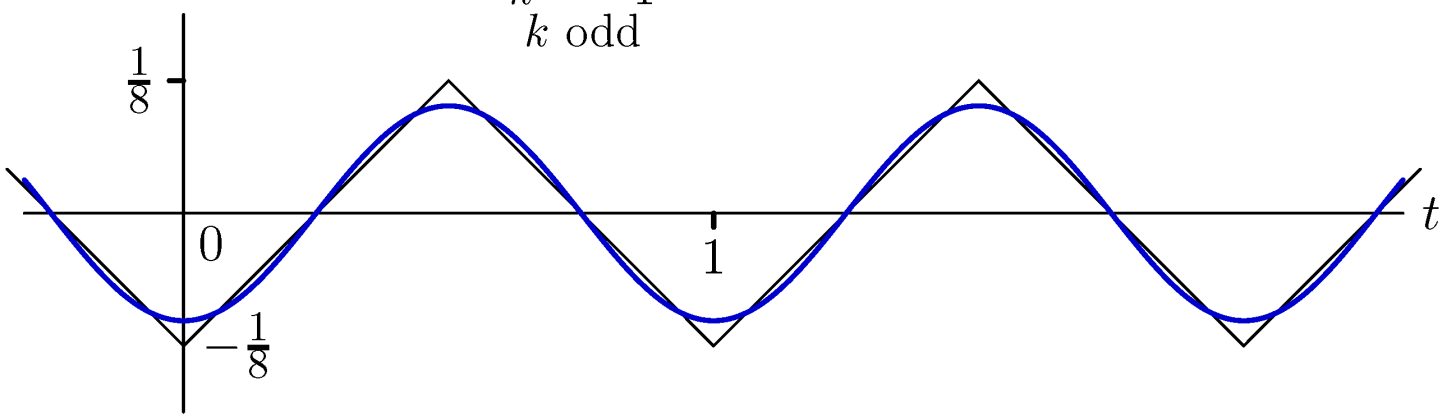


Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

Example: triangle waveform

$$\sum_{\substack{k=-1 \\ k \text{ odd}}}^{1} \frac{-1}{2k^2\pi^2} e^{j2\pi kt}$$

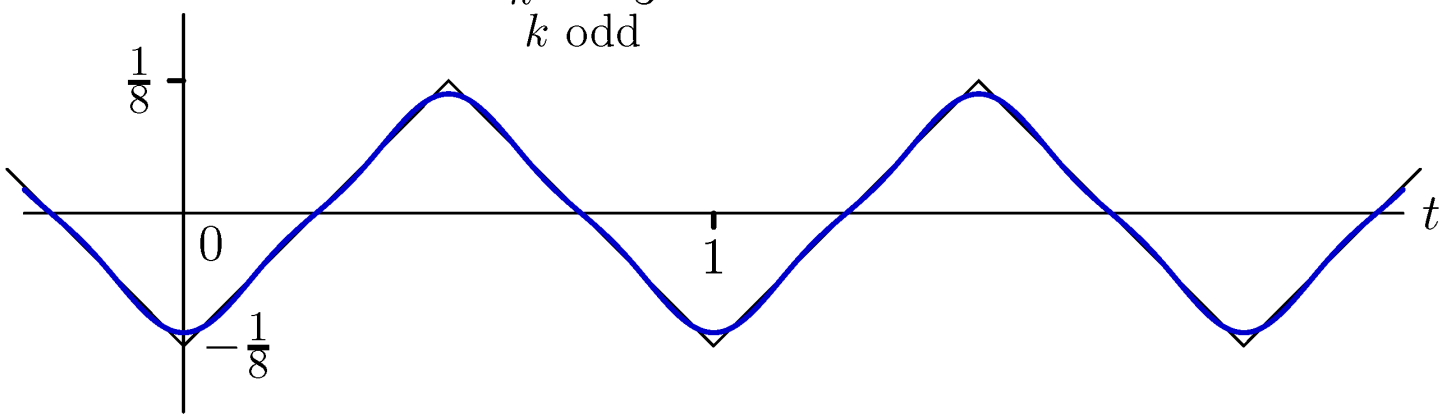


Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

Example: triangle waveform

$$\sum_{\substack{k=-3 \\ k \text{ odd}}}^{3} \frac{-1}{2k^2\pi^2} e^{j2\pi kt}$$

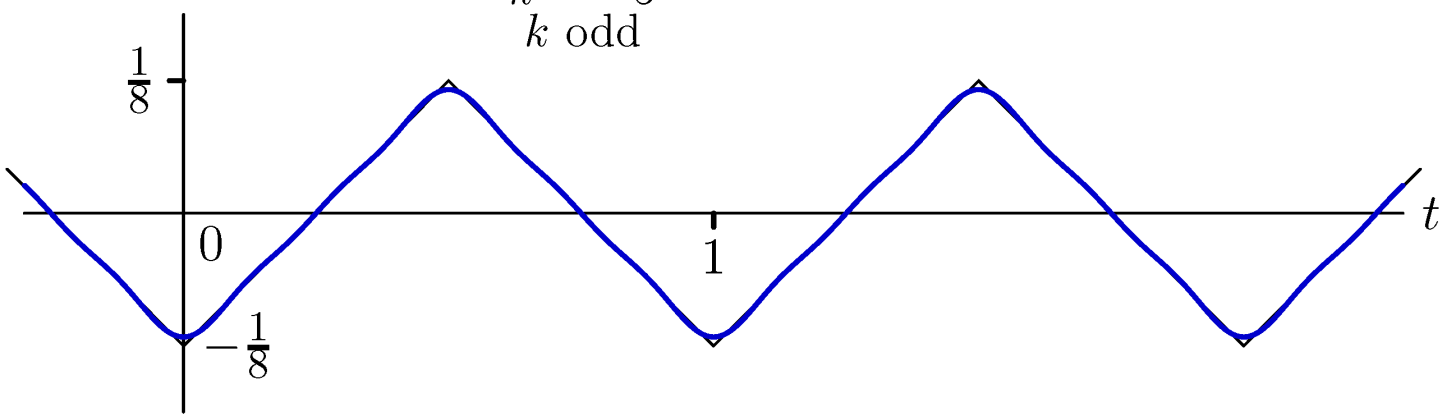


Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

Example: triangle waveform

$$\sum_{\substack{k=-5 \\ k \text{ odd}}}^{5} \frac{-1}{2k^2\pi^2} e^{j2\pi kt}$$

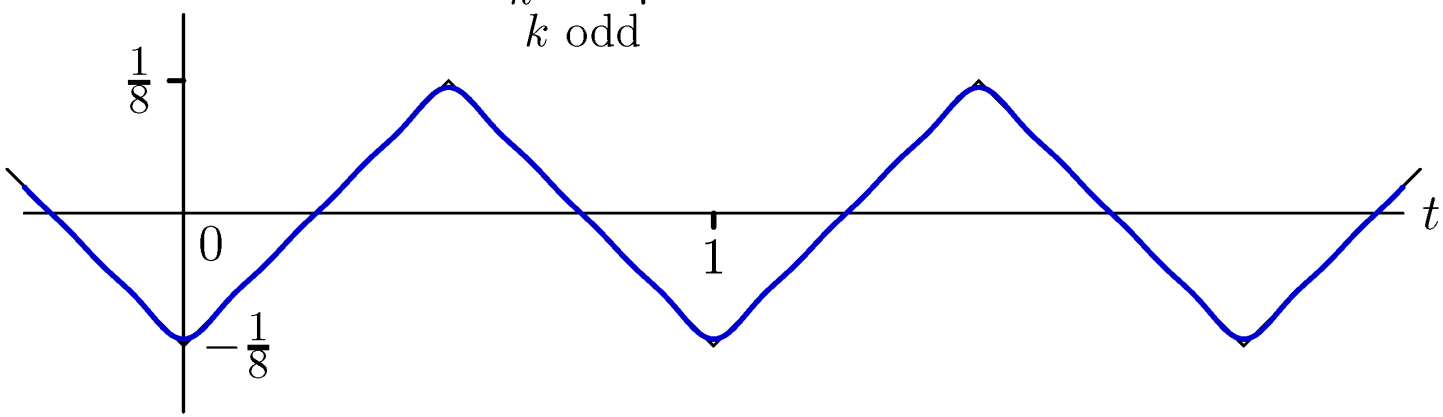


Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

Example: triangle waveform

$$\sum_{\substack{k=-7 \\ k \text{ odd}}}^{7} \frac{-1}{2k^2\pi^2} e^{j2\pi kt}$$

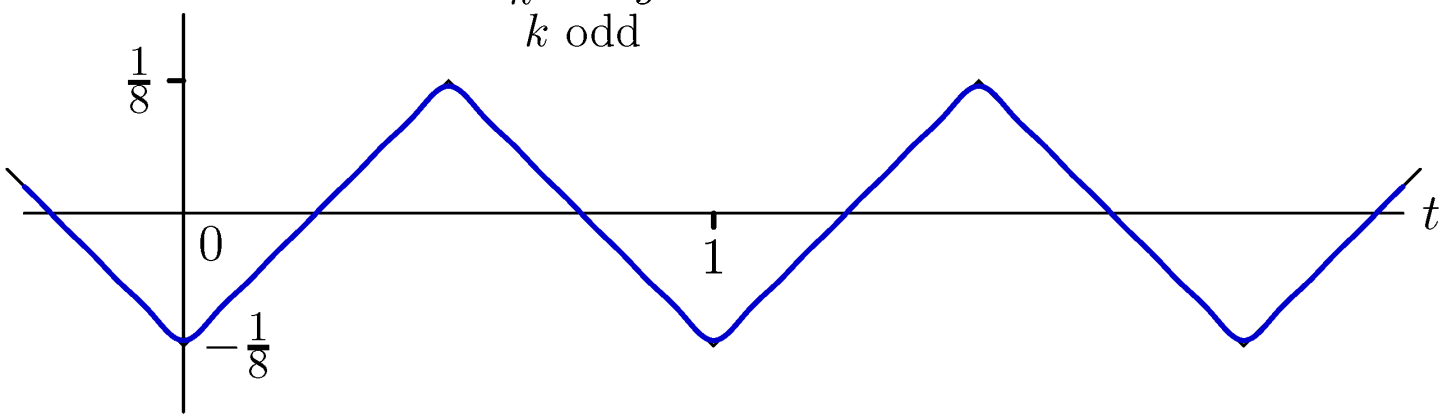


Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

Example: triangle waveform

$$\sum_{\substack{k=-9 \\ k \text{ odd}}}^{9} \frac{-1}{2k^2\pi^2} e^{j2\pi kt}$$

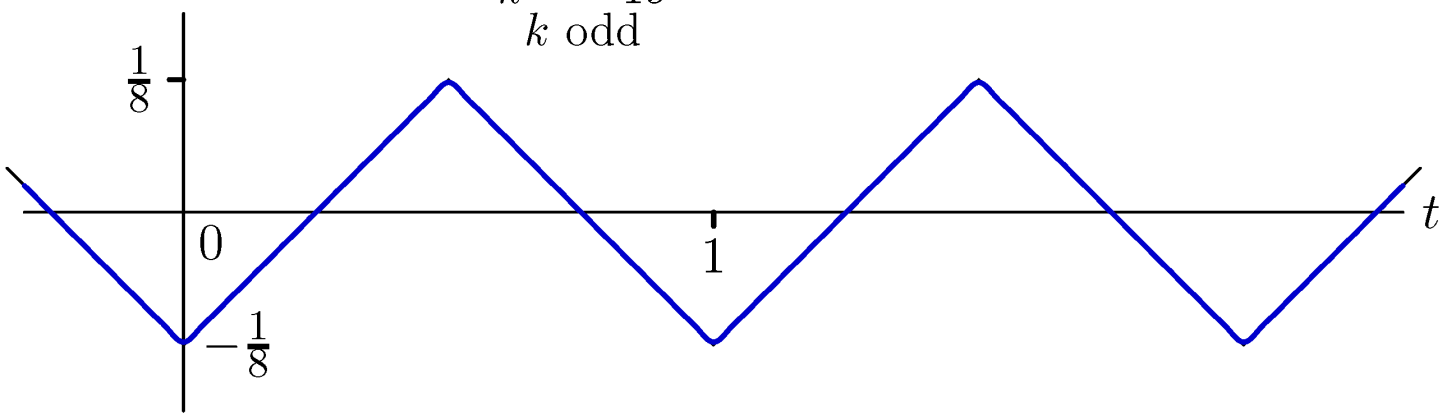


Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

Example: triangle waveform

$$\sum_{\substack{k=-19 \\ k \text{ odd}}}^{19} \frac{-1}{2k^2\pi^2} e^{j2\pi kt}$$

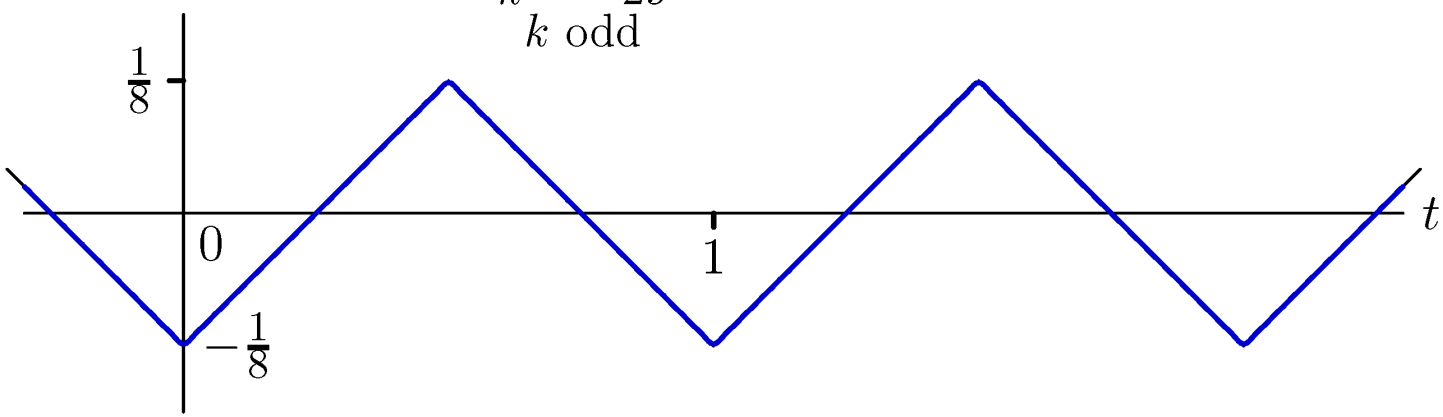


Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

Example: triangle waveform

$$\sum_{\substack{k=-29 \\ k \text{ odd}}}^{29} \frac{-1}{2k^2\pi^2} e^{j2\pi kt}$$

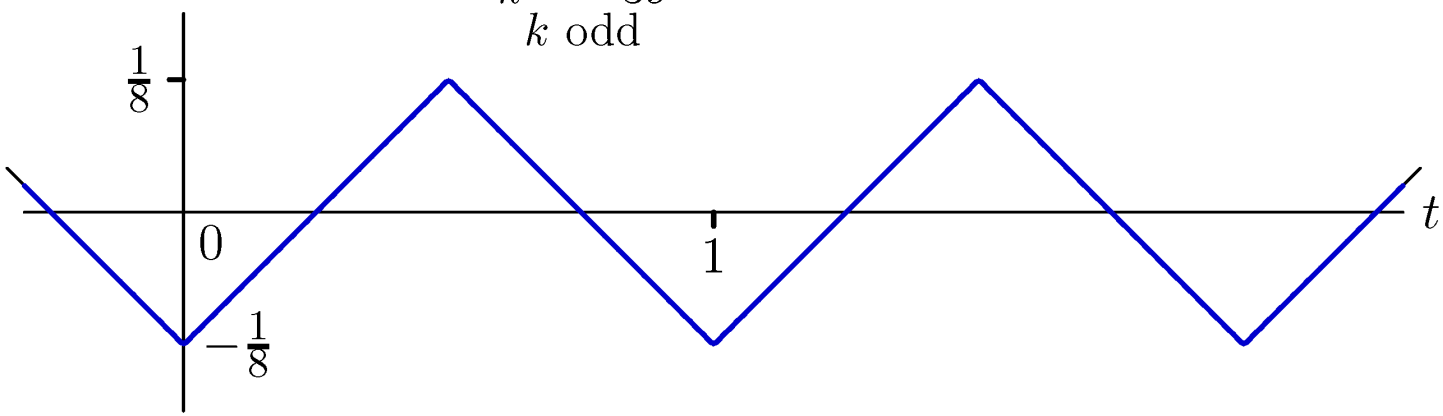


Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

Example: triangle waveform

$$\sum_{\substack{k=-39 \\ k \text{ odd}}}^{39} \frac{-1}{2k^2\pi^2} e^{j2\pi kt}$$



Fourier series representations of functions with discontinuous slopes converge toward functions with discontinuous slopes.

Fourier Series

The previous example illustrates that the sum of an infinite number of sinusoids can generate a piecewise linear function **with discontinuous slope**!

Note that **none** of the Fourier components had a slope discontinuity.

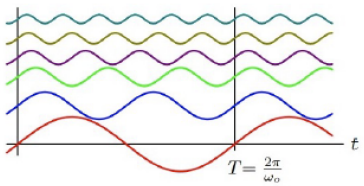
What if the function of interest has **discontinuous values**?

Fourier Series

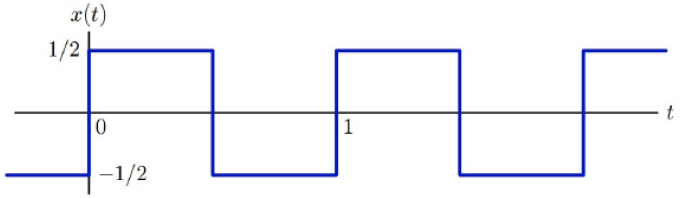
A debate two hundred years ago…

$$f(t) = c_0 + \sum_{k=1}^{\infty} \left(c_k \cos(k\omega_o t) + d_k \sin(k\omega_o t) \right)$$

Q: What about discontinuous functions?



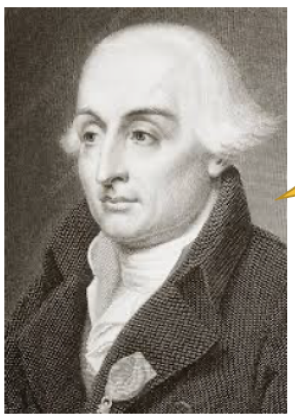
Continuous → discontinuous ?



Not a problem



Jean-Baptiste Joseph Fourier



Joseph-Louis Lagrange

No way

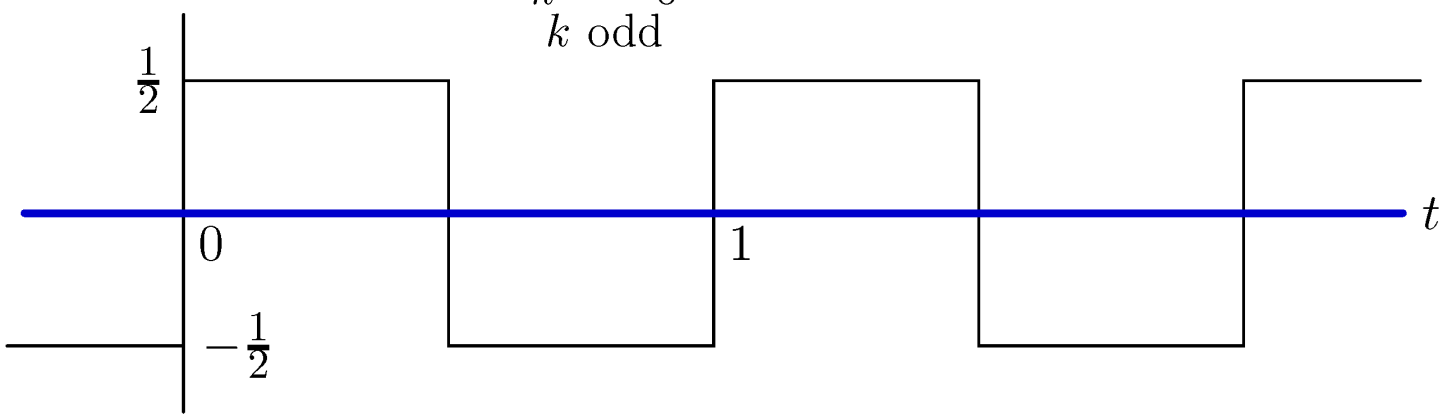
Q: What do you think?

Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

Example: square wave

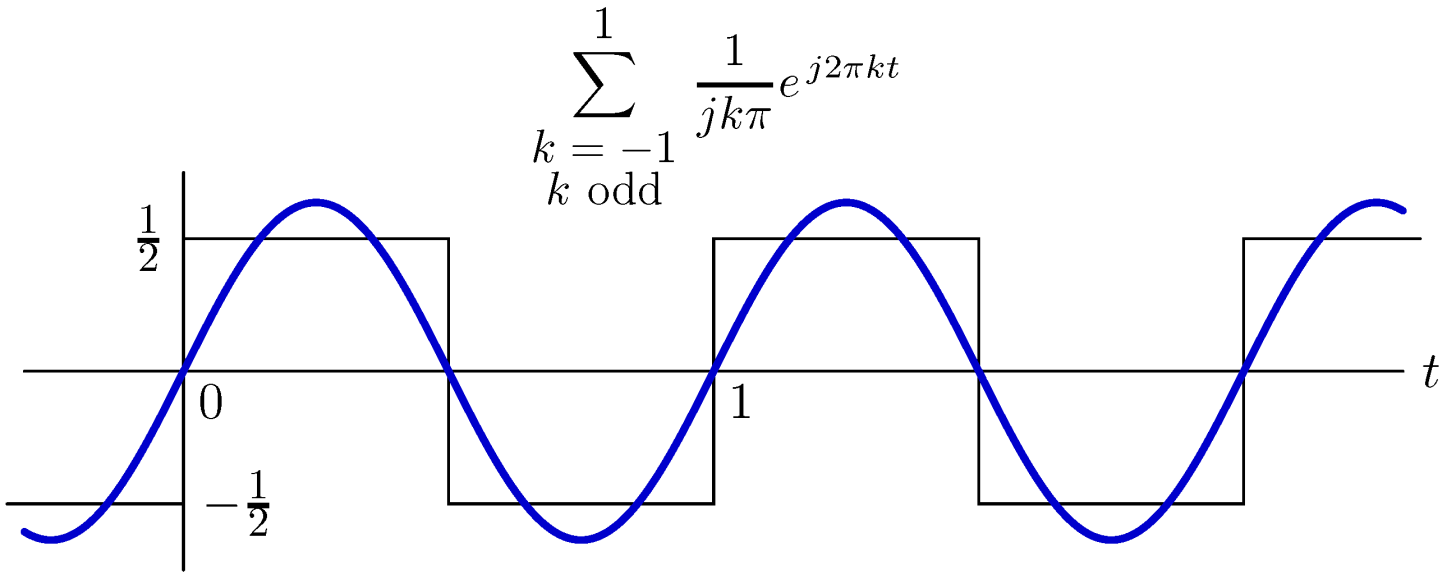
$$\sum_{\substack{k=-0 \\ k \text{ odd}}}^{0} \frac{1}{jk\pi} e^{j2\pi kt}$$



Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

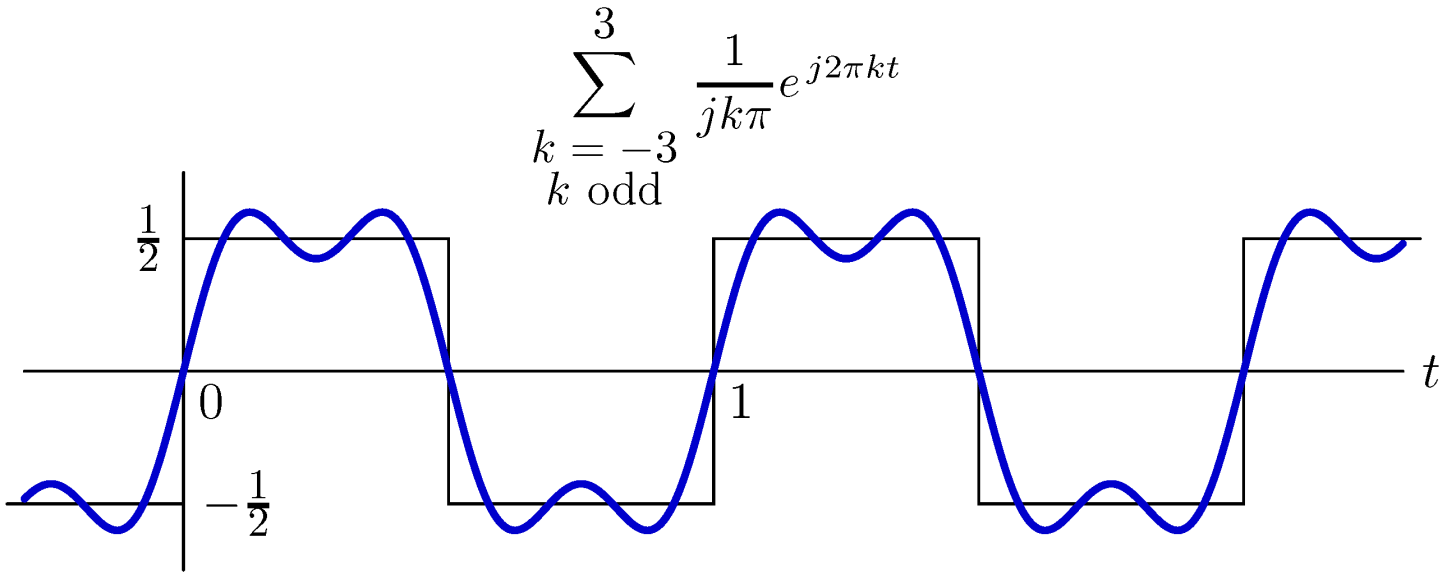
Example: square wave



Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

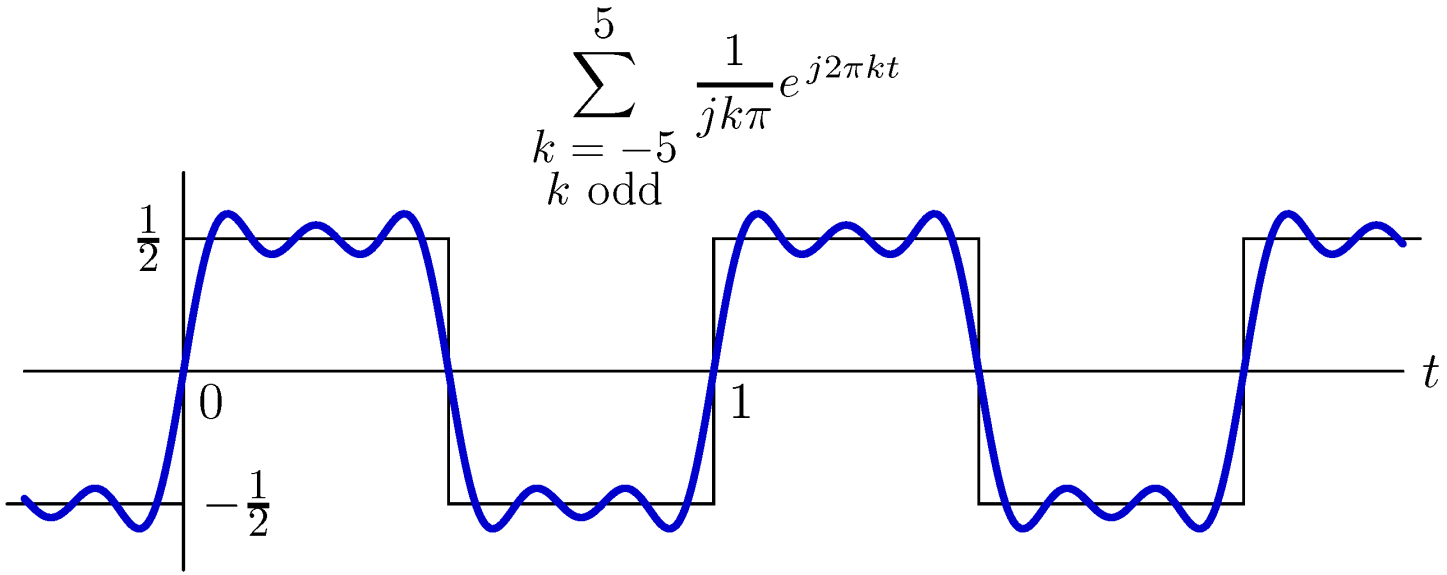
Example: square wave



Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

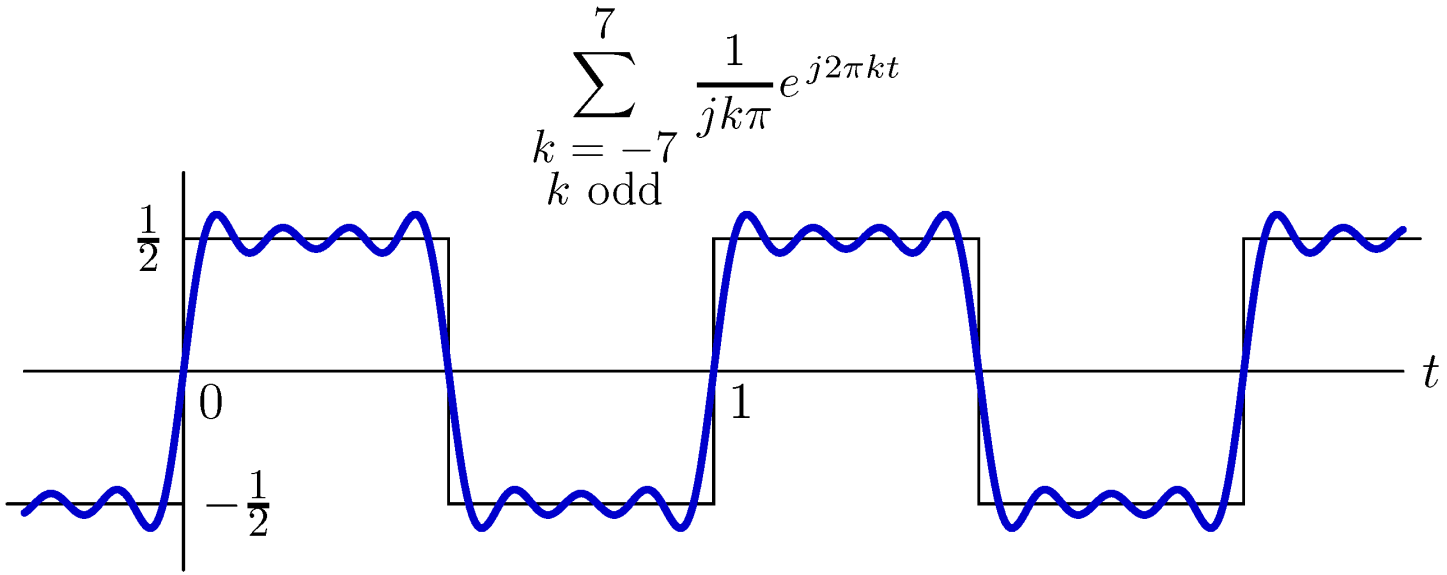
Example: square wave



Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

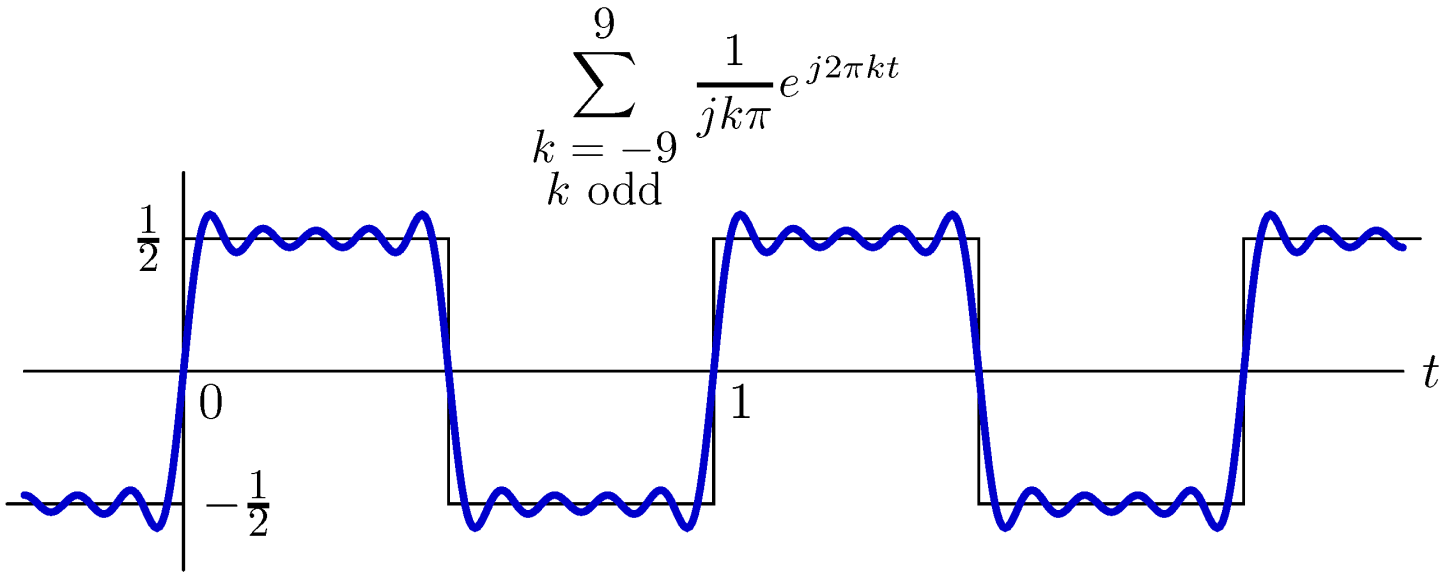
Example: square wave



Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

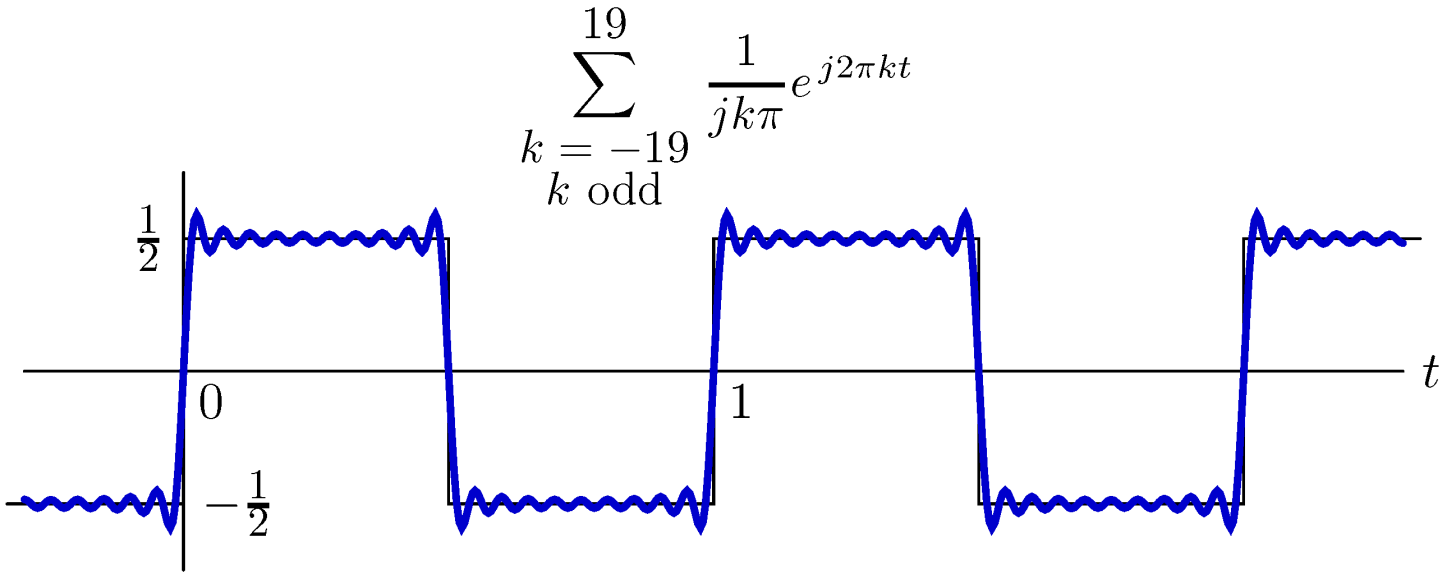
Example: square wave



Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

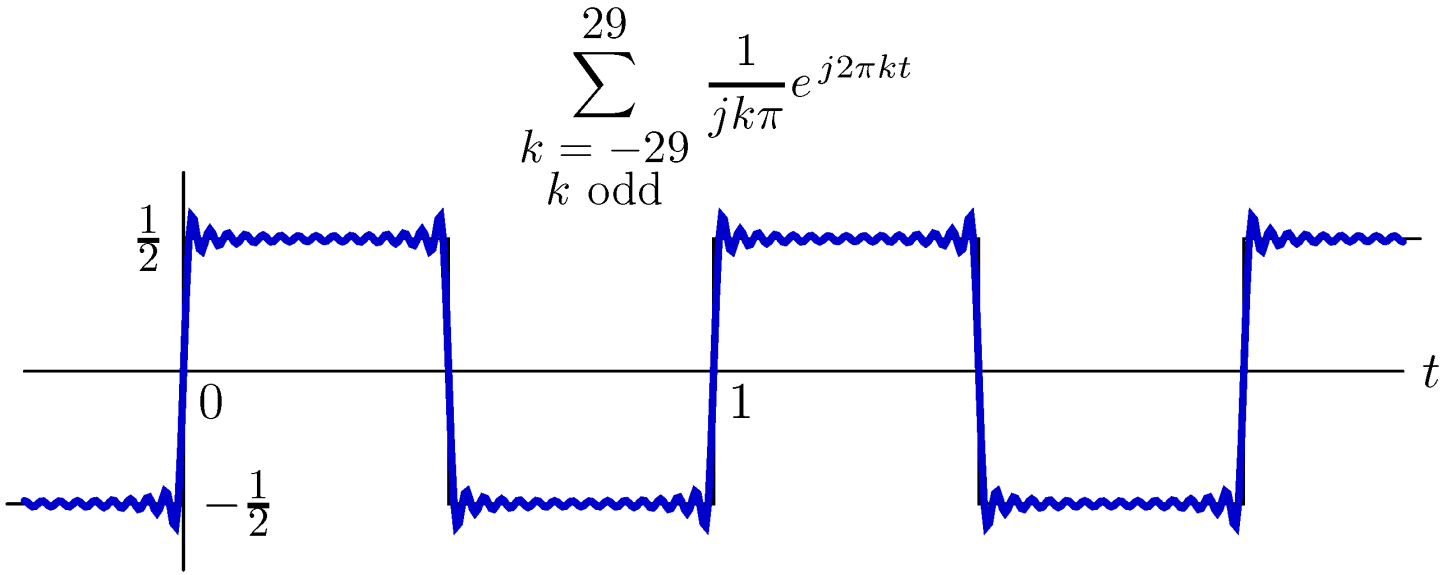
Example: square wave



Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

Example: square wave

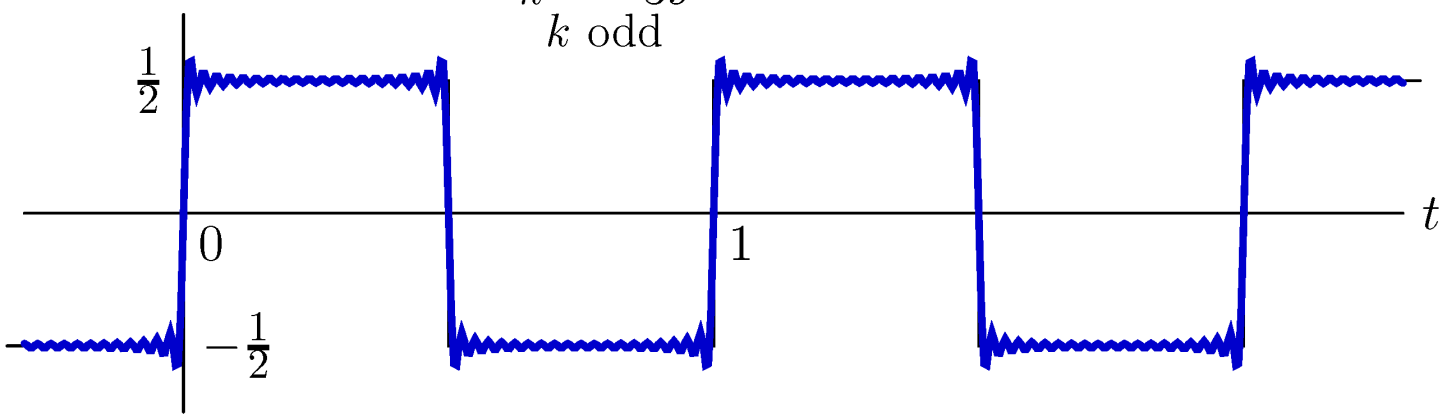


Fourier Series

One can visualize convergence of the Fourier Series by incrementally adding terms.

Example: square wave

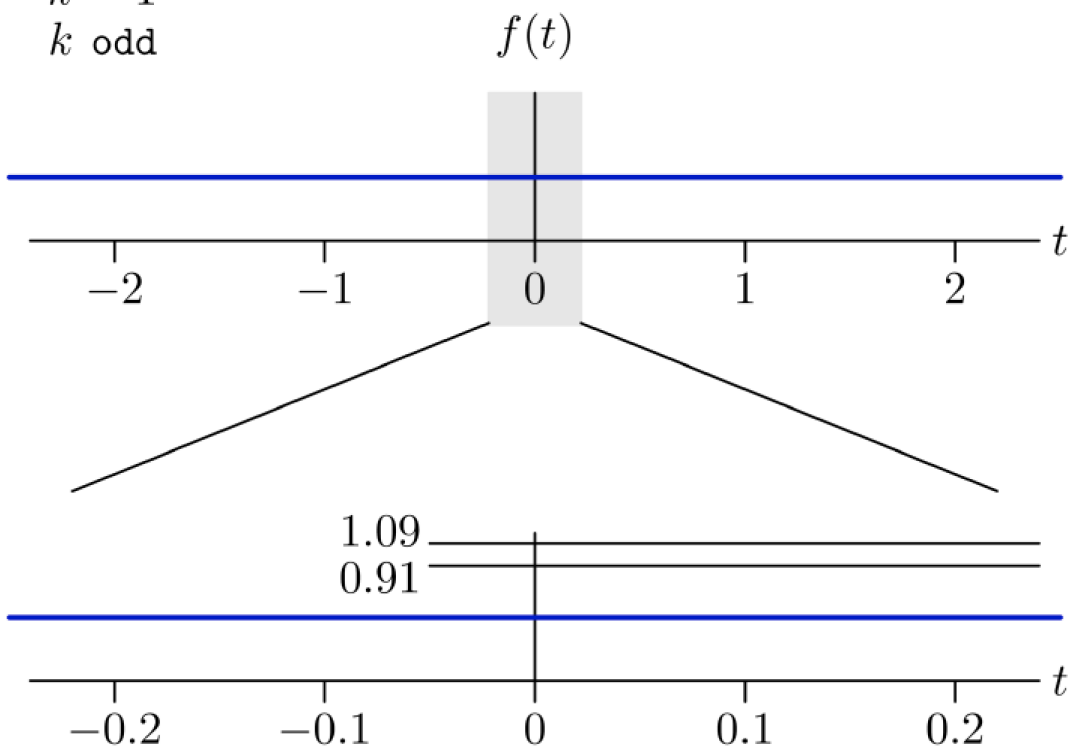
$$\sum_{\substack{k=-39 \\ k \text{ odd}}}^{39} \frac{1}{jk\pi} e^{j2\pi kt}$$



Fourier Synthesis of a Square Wave

Zoom in on the step discontinuity at $t = 0$.

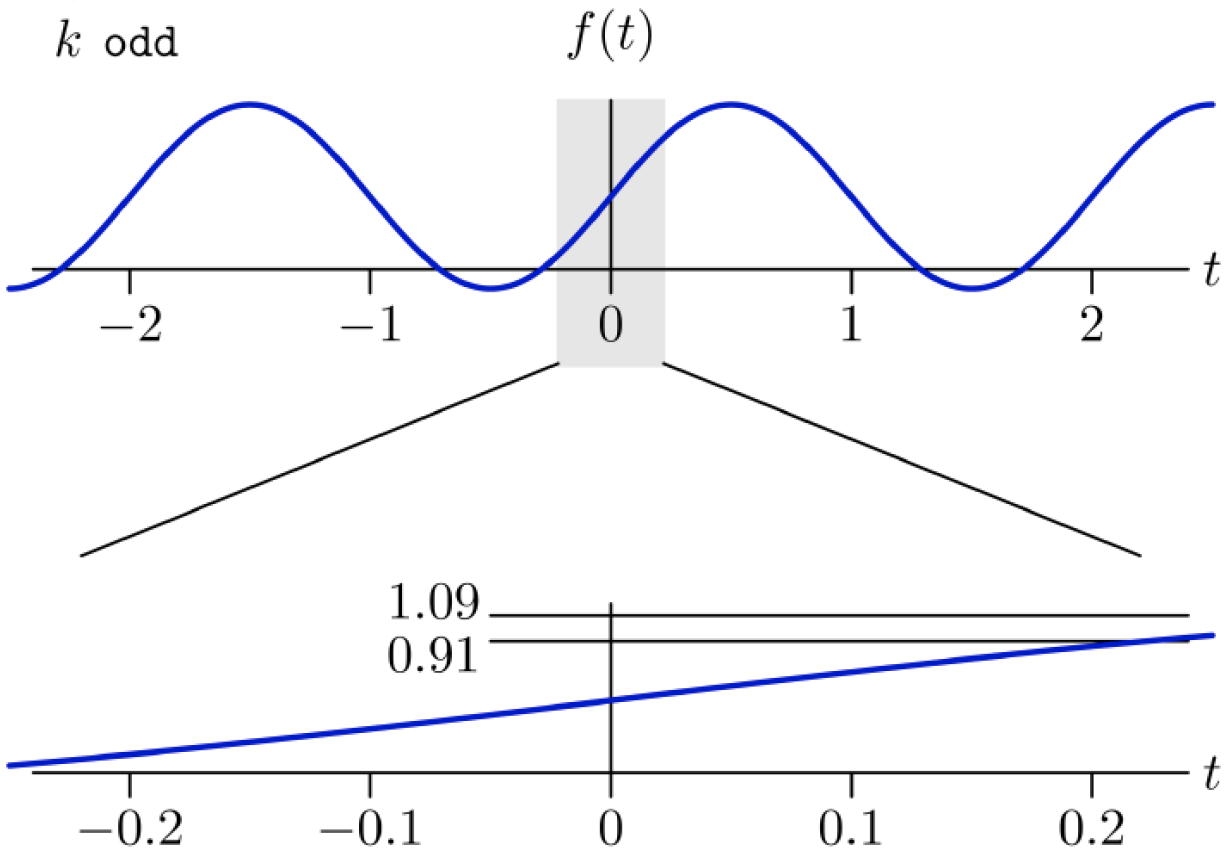
$$f(t) = \frac{1}{2} + \sum_{\substack{k=1 \\ k \text{ odd}}}^{0} \frac{2}{k\pi} \sin(k\pi t)$$



Fourier Synthesis of a Square Wave

Zoom in on the step discontinuity at $t = 0$.

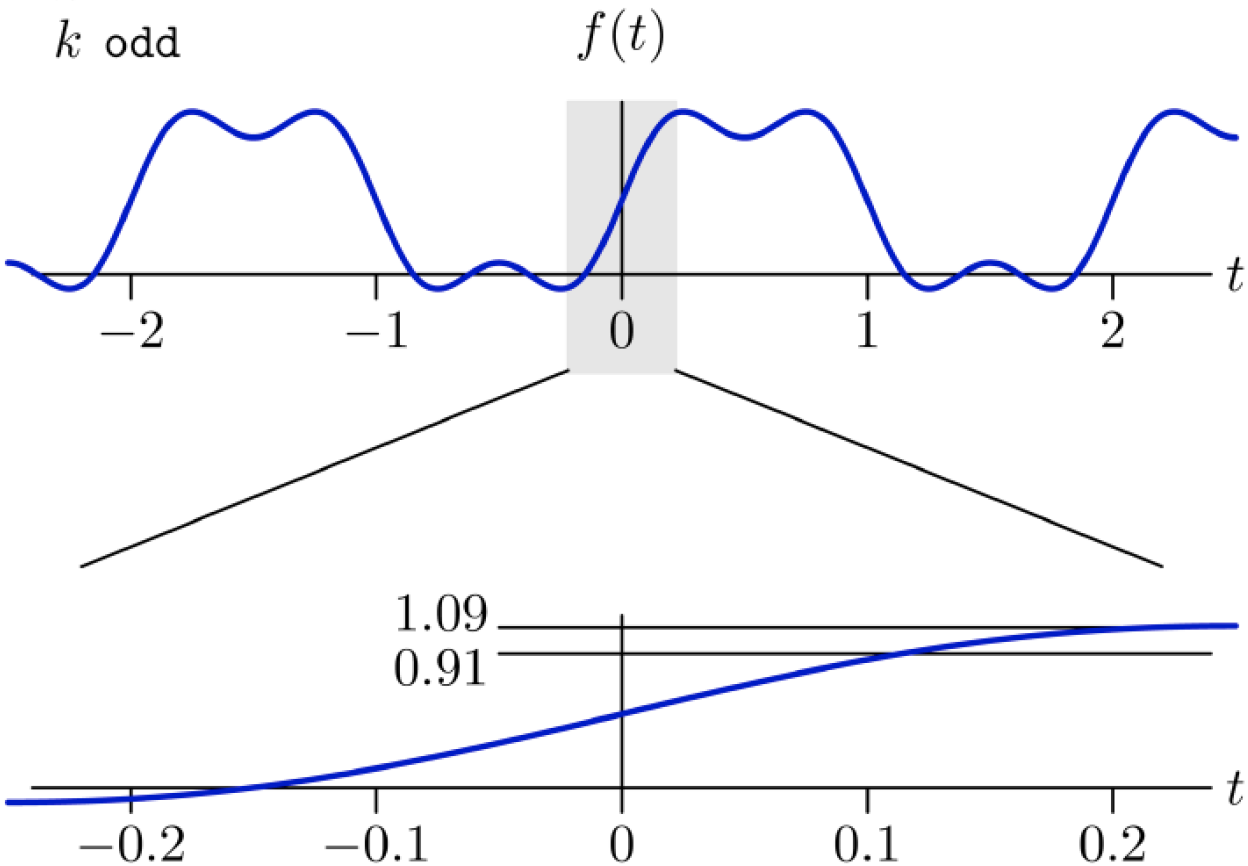
$$f(t) = \frac{1}{2} + \sum_{\substack{k=1 \\ k \text{ odd}}}^{1} \frac{2}{k\pi} \sin(k\pi t)$$



Fourier Synthesis of a Square Wave

Zoom in on the step discontinuity at $t = 0$.

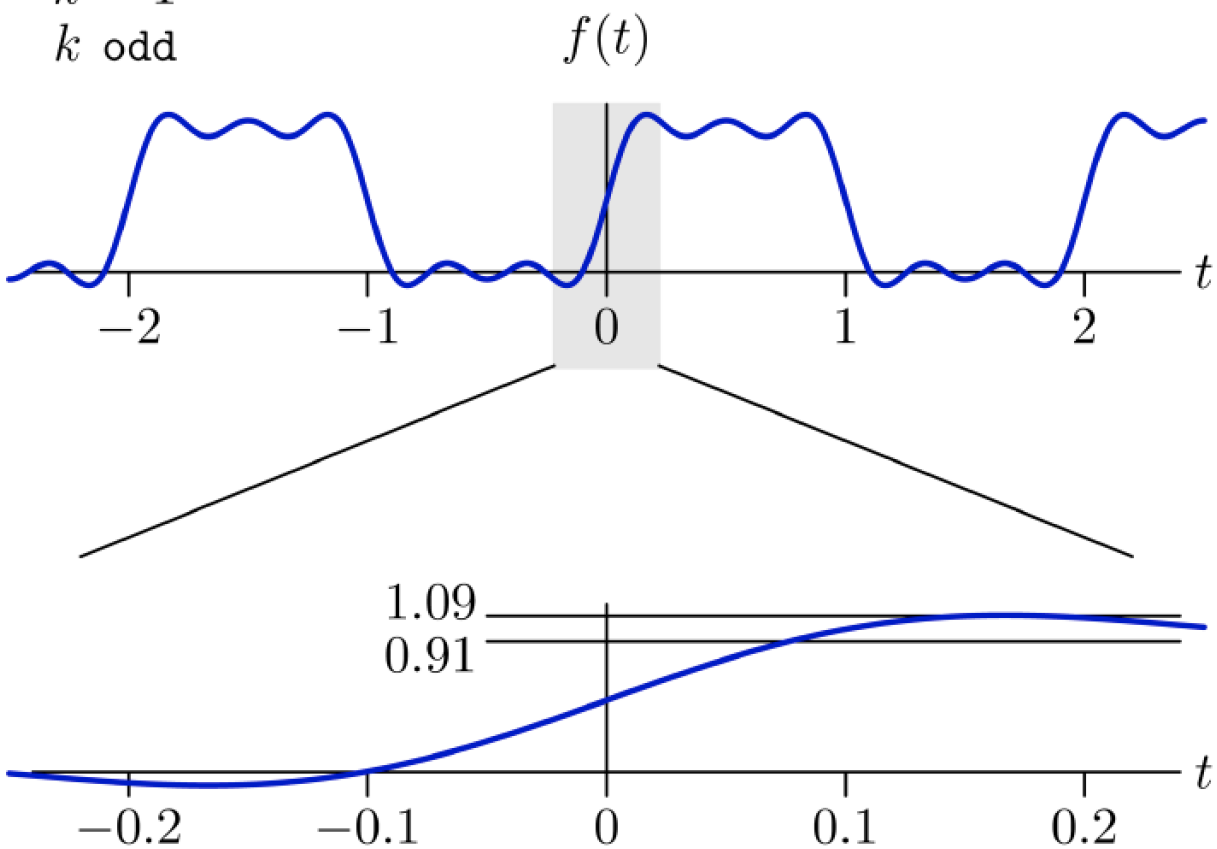
$$f(t) = \frac{1}{2} + \sum_{\substack{k=1 \\ k \text{ odd}}}^{3} \frac{2}{k\pi} \sin(k\pi t)$$



Fourier Synthesis of a Square Wave

Zoom in on the step discontinuity at $t = 0$.

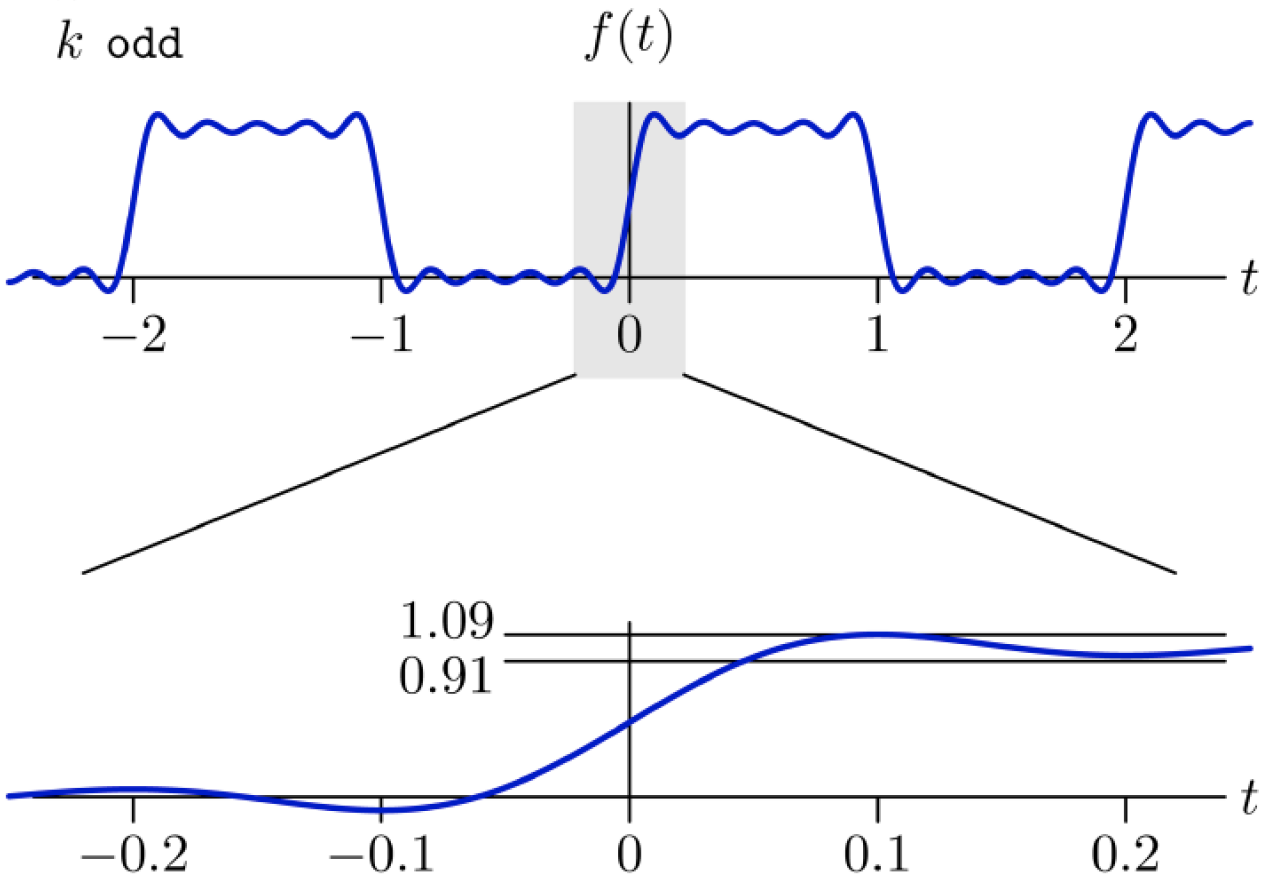
$$f(t) = \frac{1}{2} + \sum_{\substack{k=1 \\ k \text{ odd}}}^{5} \frac{2}{k\pi} \sin(k\pi t)$$



Fourier Synthesis of a Square Wave

Zoom in on the step discontinuity at $t = 0$.

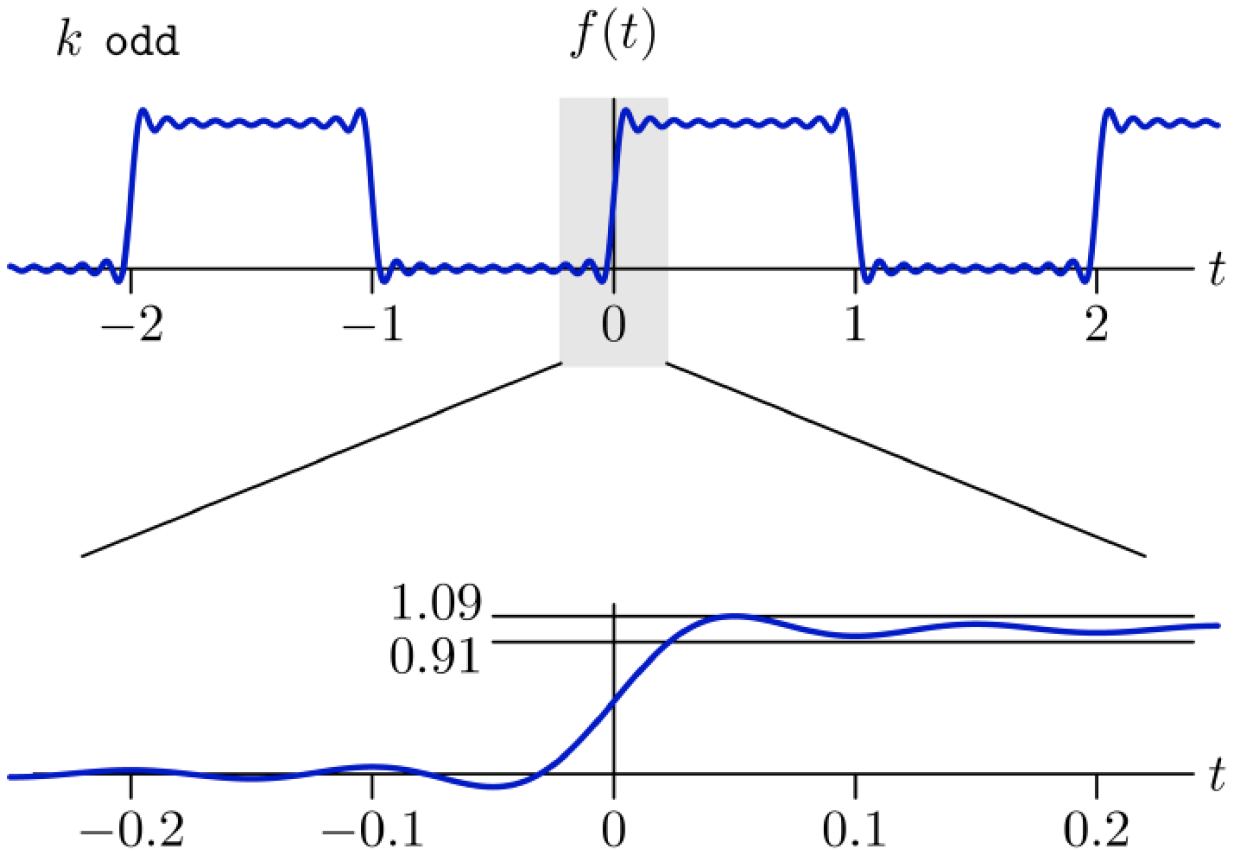
$$f(t) = \frac{1}{2} + \sum_{\substack{k=1 \\ k \text{ odd}}}^{9} \frac{2}{k\pi} \sin(k\pi t)$$



Fourier Synthesis of a Square Wave

Zoom in on the step discontinuity at $t = 0$.

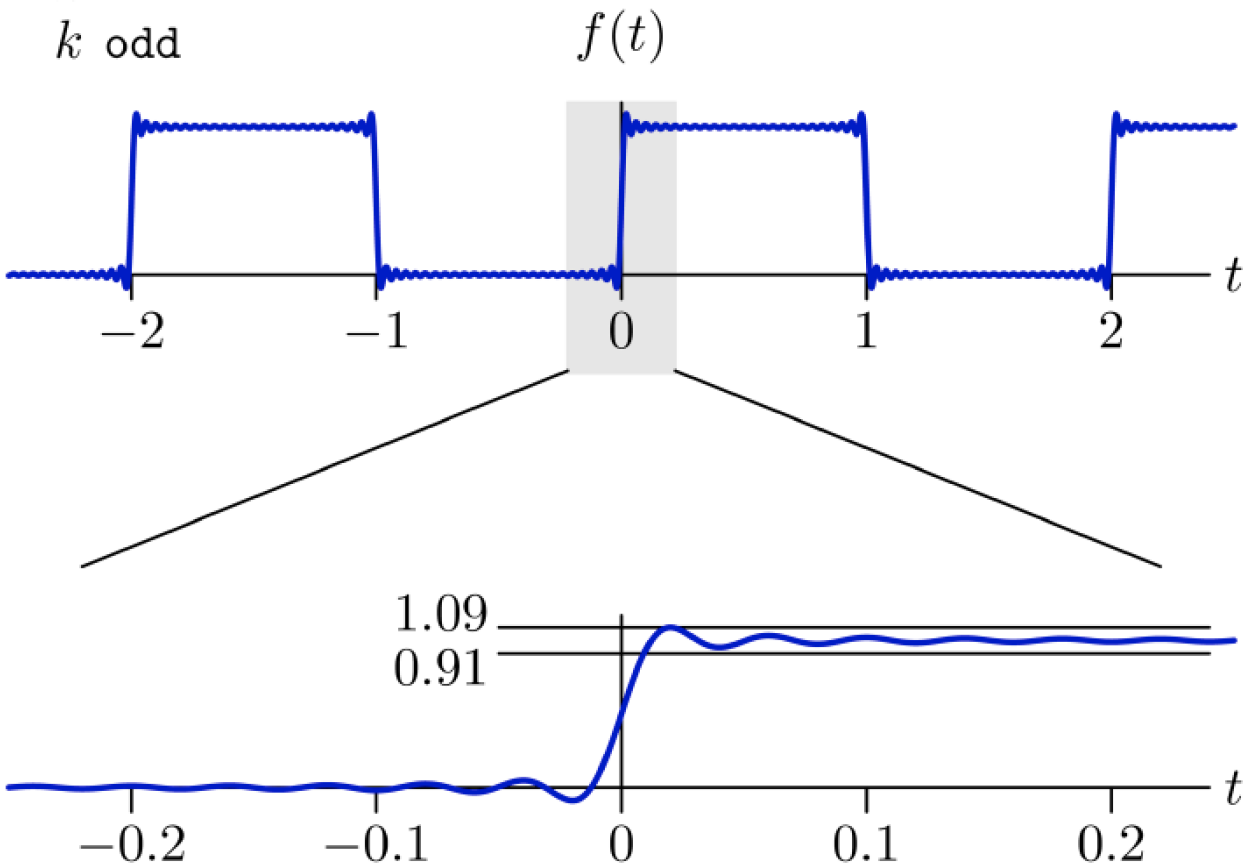
$$f(t) = \frac{1}{2} + \sum_{\substack{k=1 \\ k \text{ odd}}}^{19} \frac{2}{k\pi} \sin(k\pi t)$$



Fourier Synthesis of a Square Wave

Zoom in on the step discontinuity at $t = 0$.

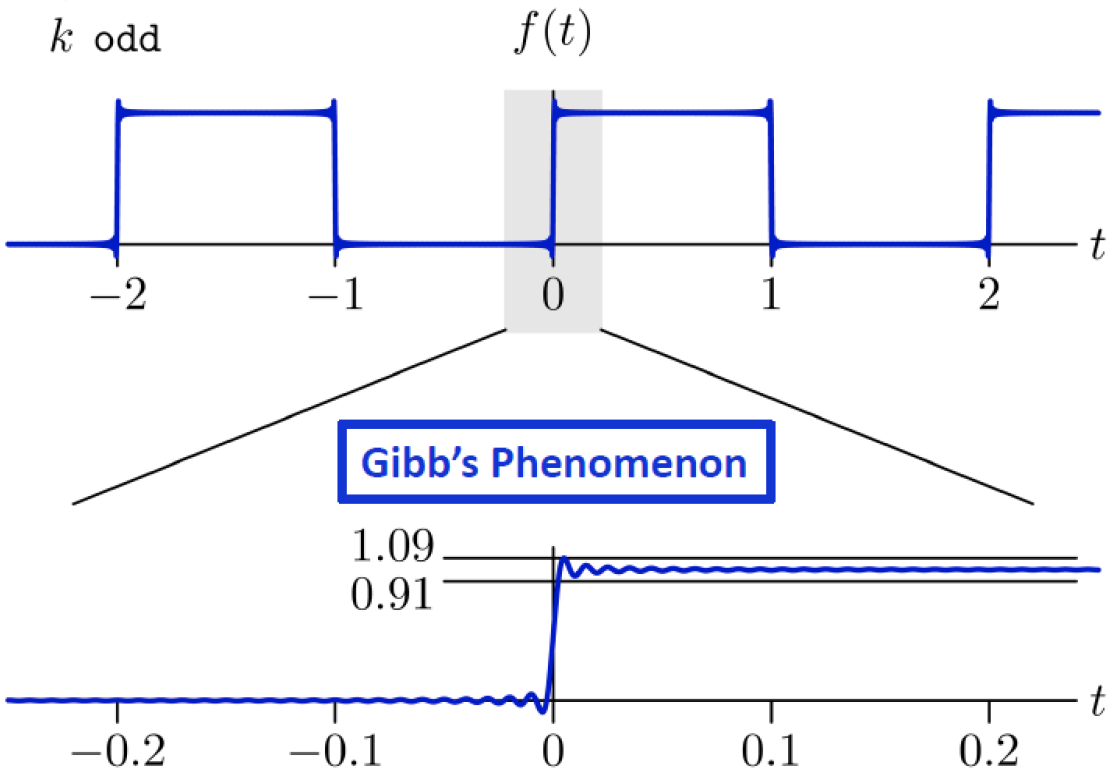
$$f(t) = \frac{1}{2} + \sum_{\substack{k=1 \\ k \text{ odd}}}^{49} \frac{2}{k\pi} \sin(k\pi t)$$



Fourier Synthesis of a Square Wave

Zoom in on the step discontinuity at $t = 0$.

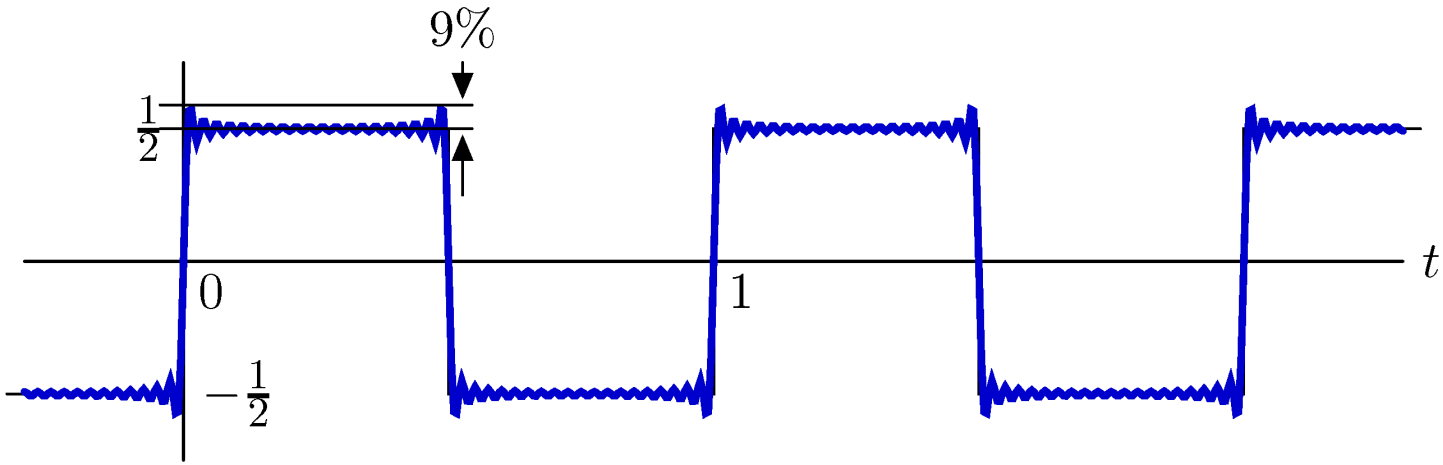
$$f(t) = \frac{1}{2} + \sum_{\substack{k=1 \\ k \text{ odd}}}^{199} \frac{2}{k\pi} \sin(k\pi t)$$



Increasing the number of terms does not decrease the peak overshoot, but it does shrink the region of time that is occupied by the overshoot.

Fourier Series

Partial sums of Fourier series of discontinuous functions "ring" near discontinuities: Gibb's phenomenon.



This ringing results because the magnitude of the Fourier coefficients is only decreasing as $\frac{1}{k}$ (while they decreased as $\frac{1}{k^2}$ for the triangle).

You can decrease (and even eliminate the ringing) by decreasing the magnitudes of the Fourier coefficients at higher frequencies.

Gibbs Phenomenon

$$x_N(t) = \sum_{k=-N}^{N} a_k e^{jk\omega_0 t}$$

- As $N \to \infty$, $x_N(t)$ exhibits Gibb's phenomenon at points of discontinuity;
- As N increases, the ripples in the partial sums become compressed toward the discontinuity;
- For any finite value of N, the peak amplitude of the ripples remains constant.

Fourier Series: Summary

https://realpython.com/python-scipy-fft/

Fourier series represent periodic signals as sums of sinusoids.

- valid for an extremely large class of periodic signals
- valid even for discontinuous signals such as square wave

However, convergence as # harmonics increases can be complicated.

Assignments

- Reading Assignment: Ch. 1.3, Ch. 3.0 - 3.5
- Homework 5